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Análise clúster de datos de entrada para simulacións de sistemas fotovoltaicos con afección de sombras locais

UNIVERSIDADE DA CORUÑA

USC UNIVERSIDADE DE SANTIAGO DE COMPOSTELA

Universidade Vigo

Adrián Blanco Aguiar


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Online platform for photovoltaic system simulations

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Online platform for photovoltaic system simulations

Crucial for system sizing



Online platform for photovoltaic system simulations

Crucial for system sizing

Open source simulation methods



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Online platform for photovoltaic system simulations

Crucial for system sizing

Open source simulation methods

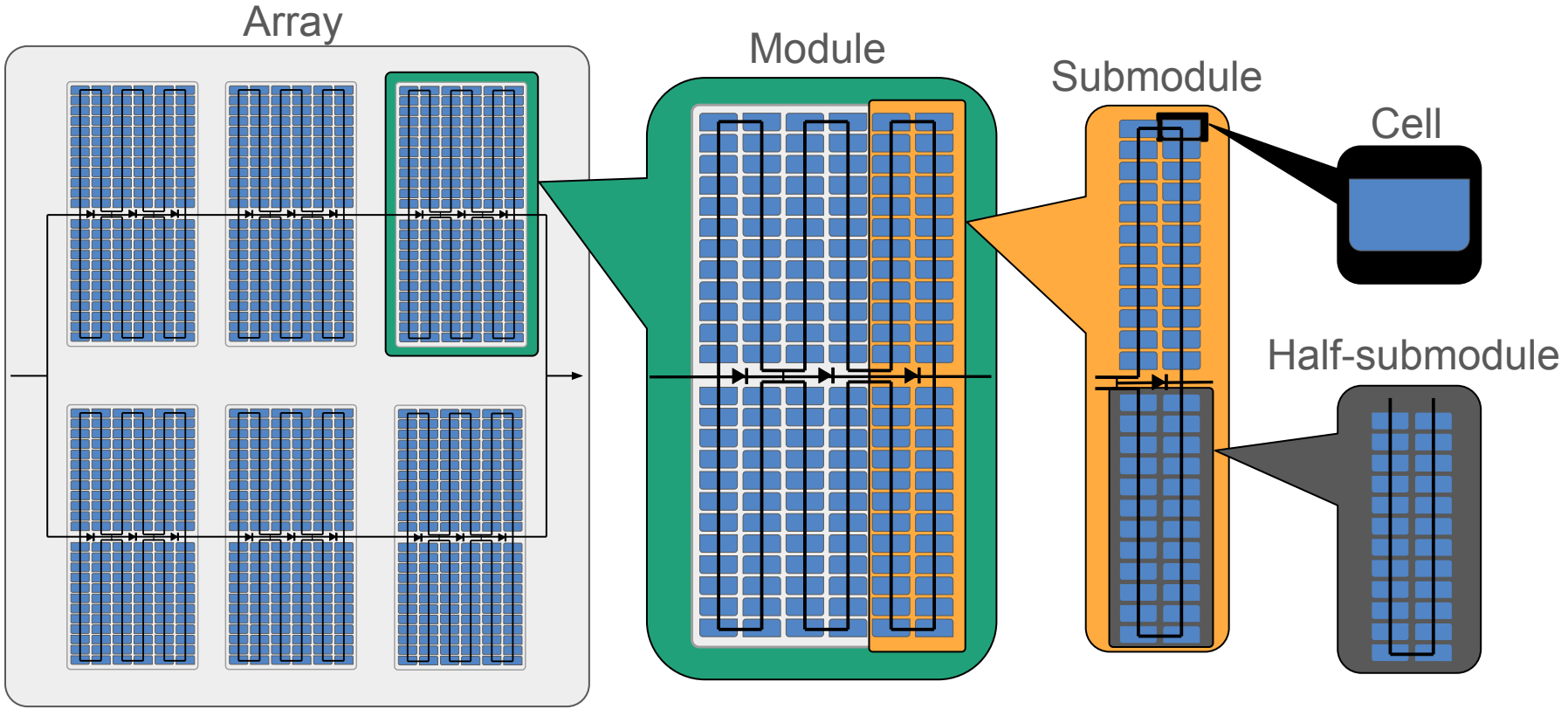
Brief photovoltaic introduction



Credits:https://unsplash.com/es/fotos/paneles-solares-en-campo-verde-513dBrMJ_5w

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Brief photovoltaic introduction



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Brief photovoltaic introduction

Cell

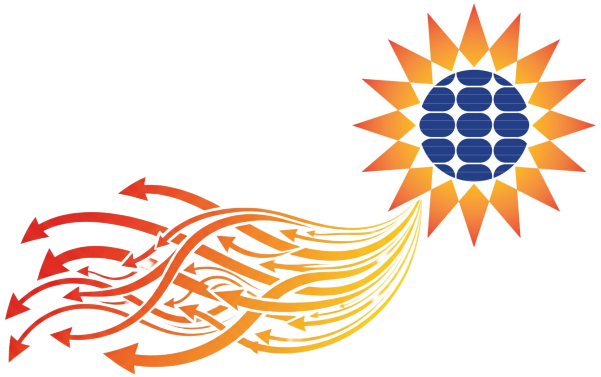


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Brief photovoltaic introduction

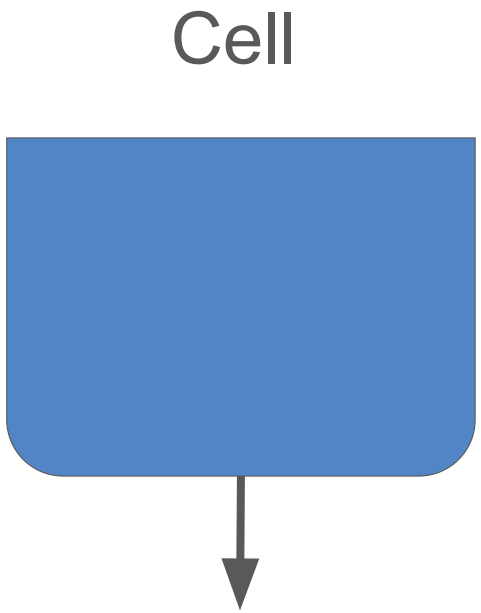
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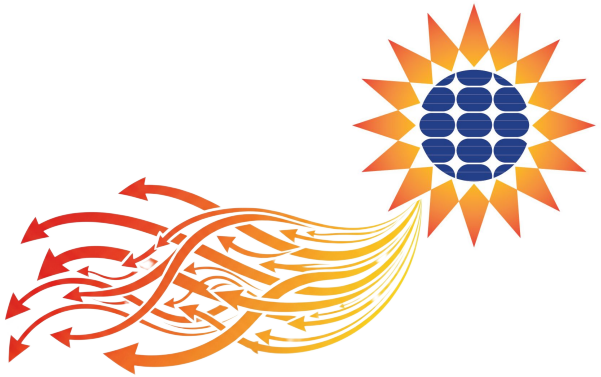




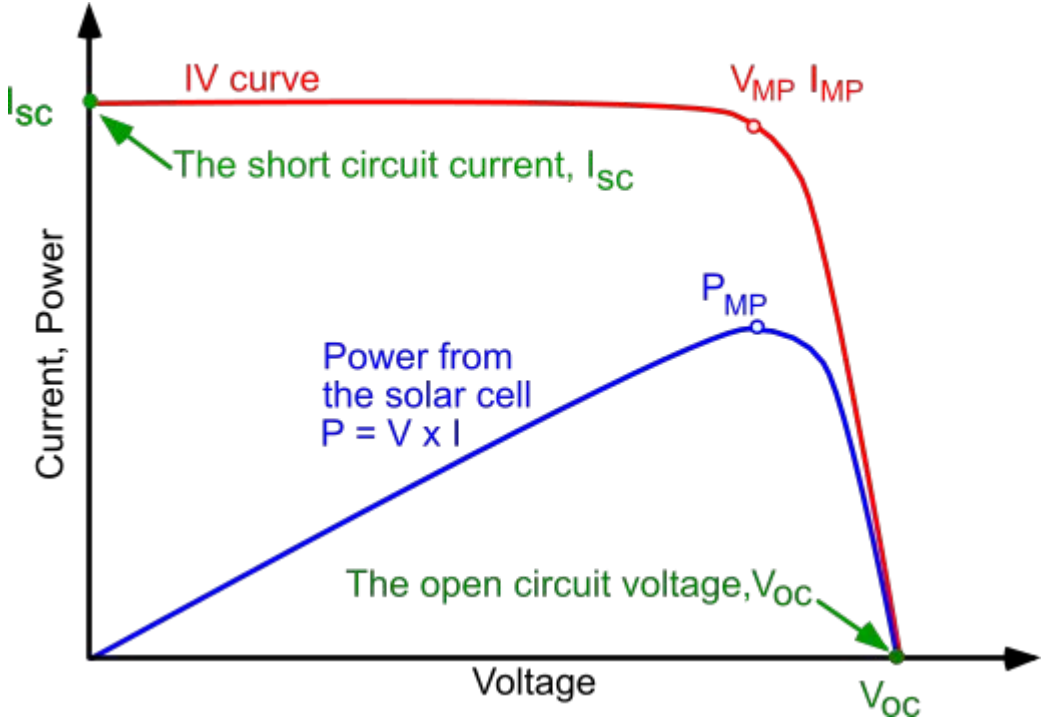
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Brief photovoltaic introduction



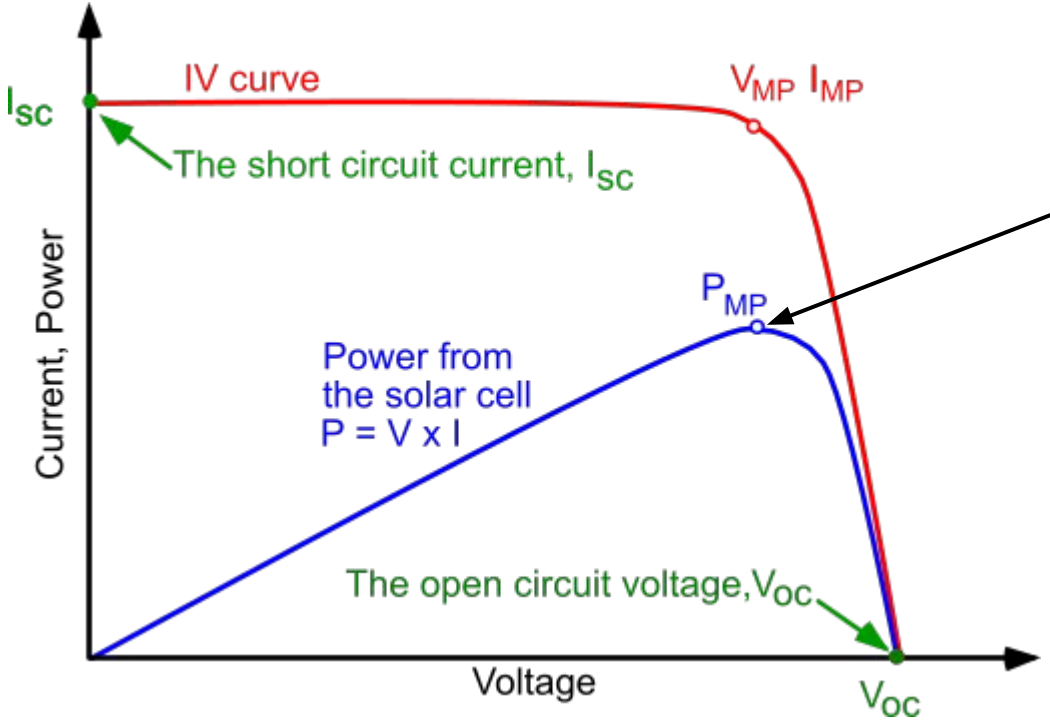


IV curve and MPP



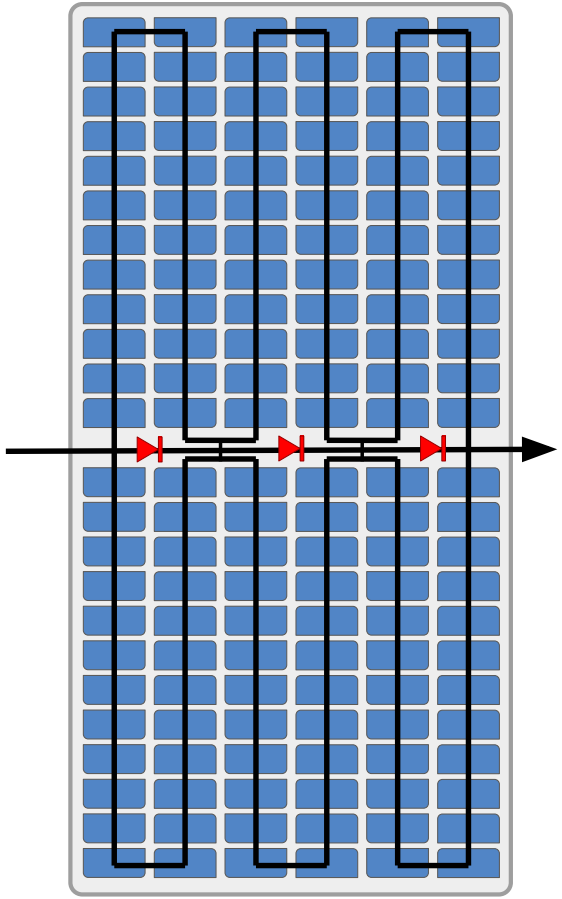
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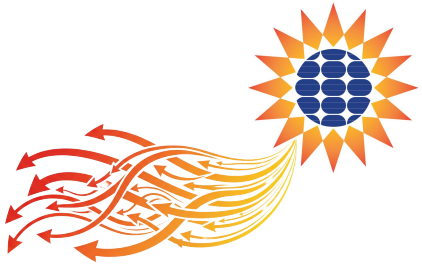
IV curve and MPP



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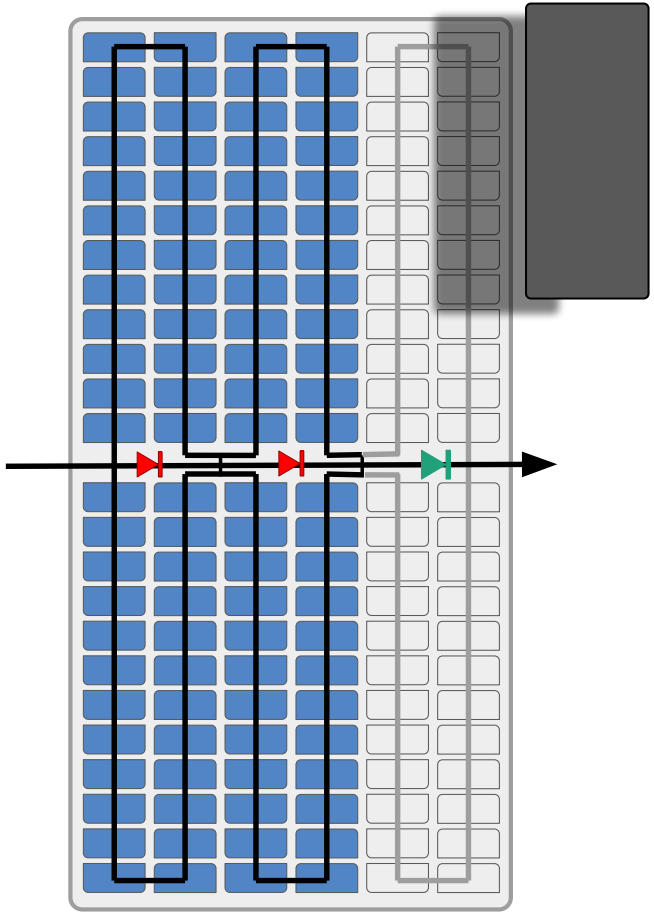
Full module

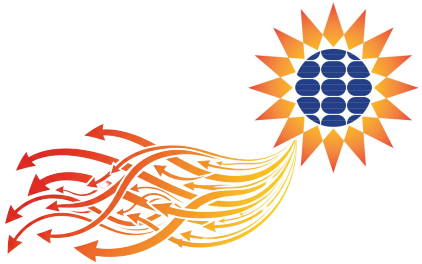




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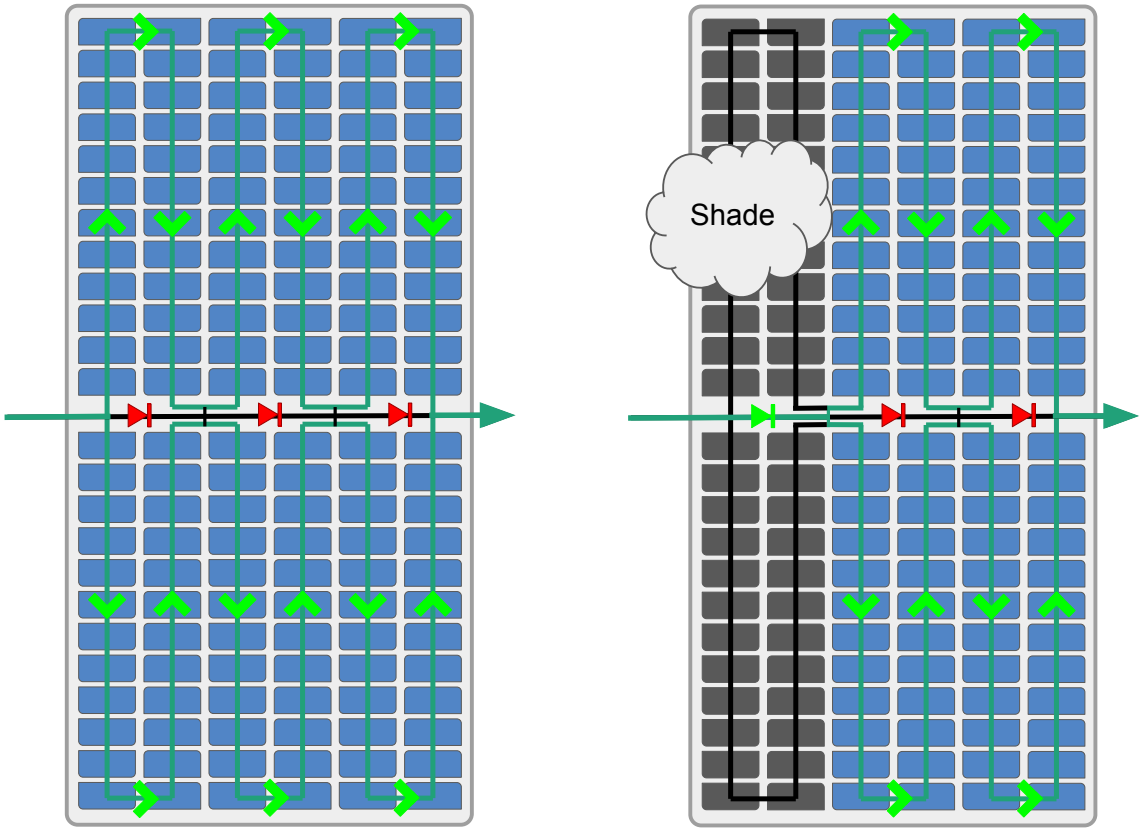
Shading mismatch





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Shading mismatch



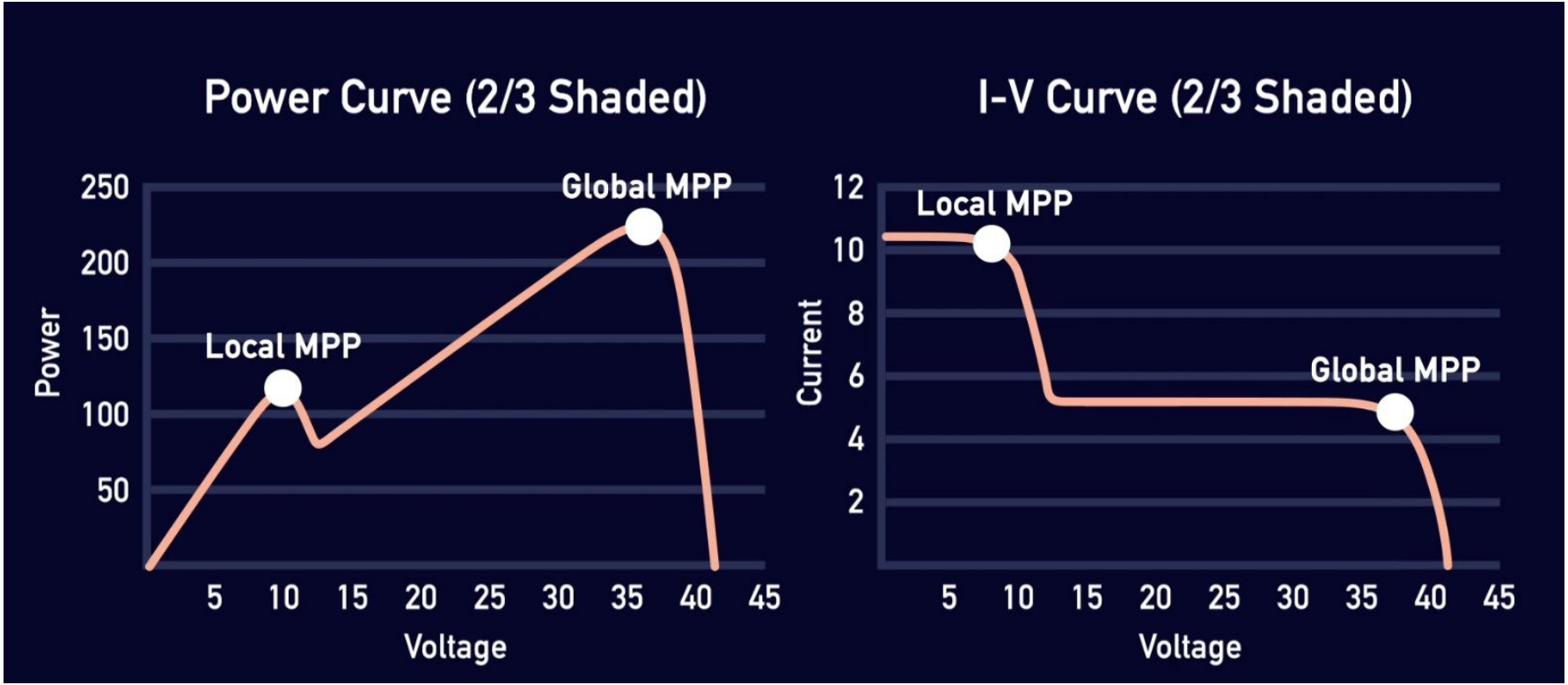
Bypass diodes

Active

Inactive

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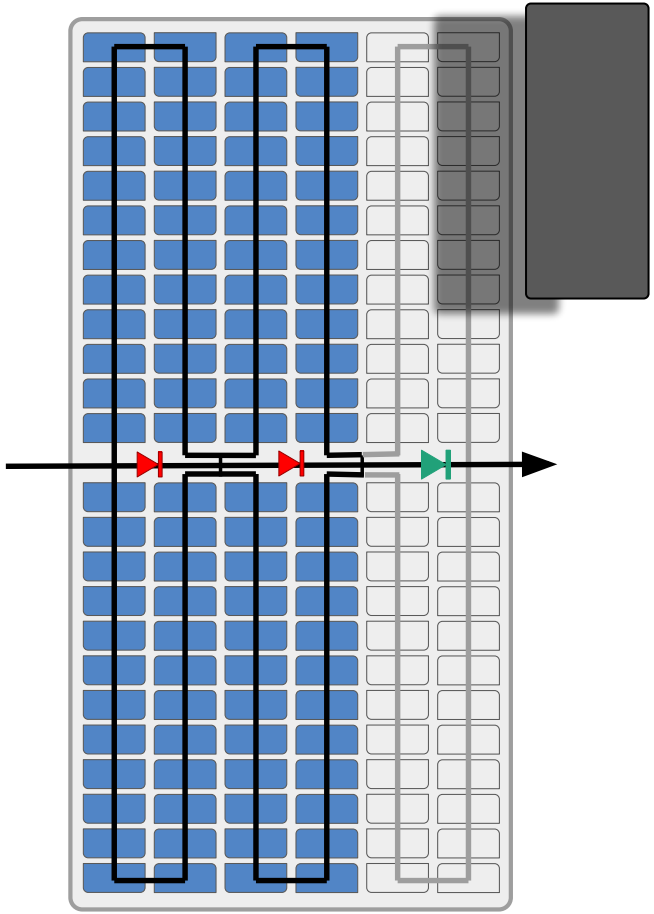
Shading mismatch

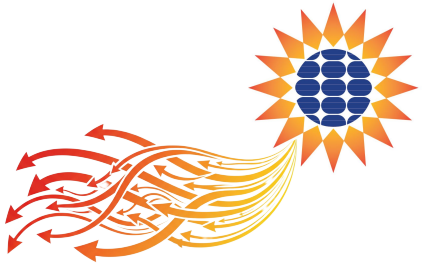


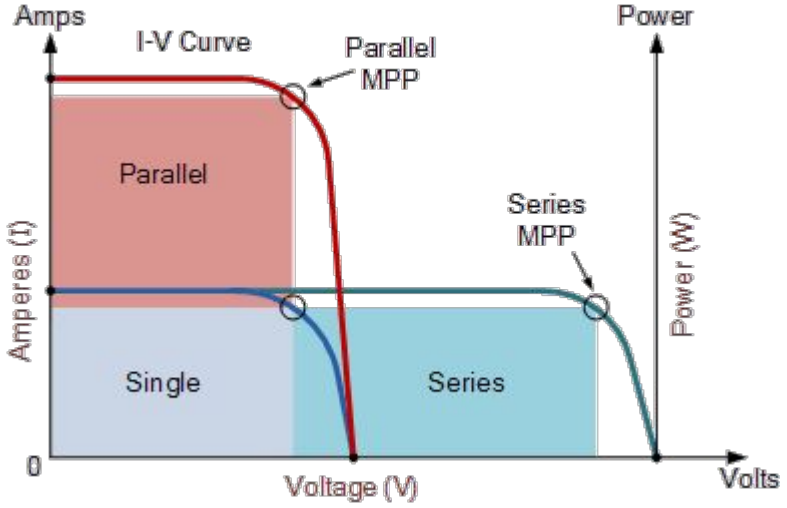
Credits: https://www.youtube.com/watch?v=AZ8VQ_zmf1M&t=209s

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Simulation process

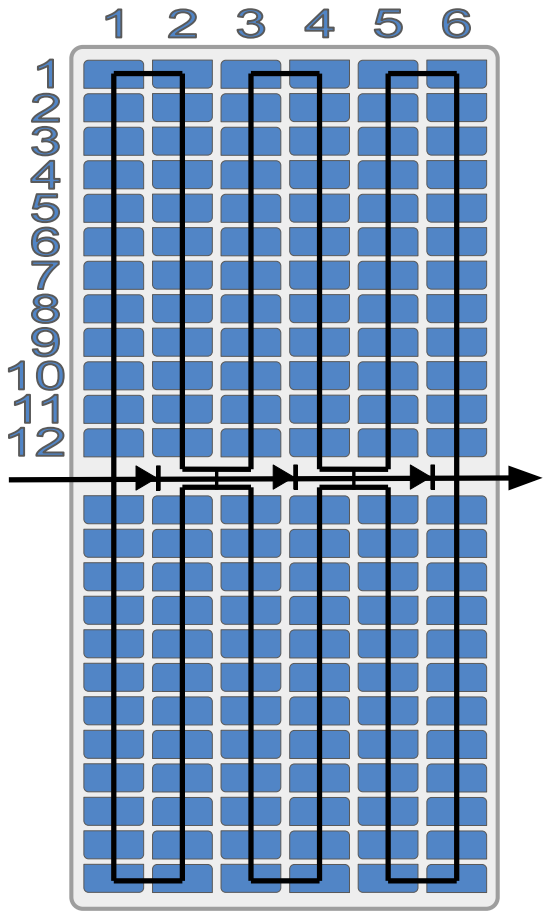






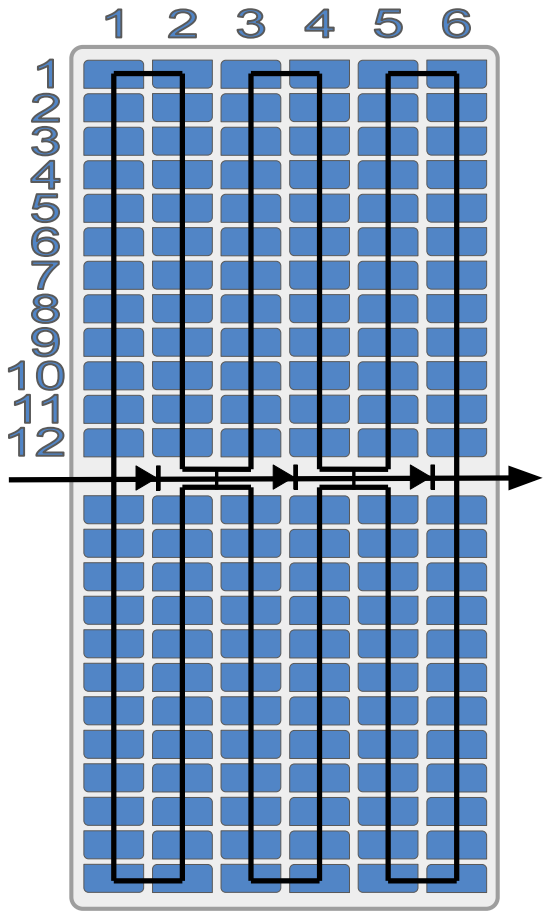
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Simulation process



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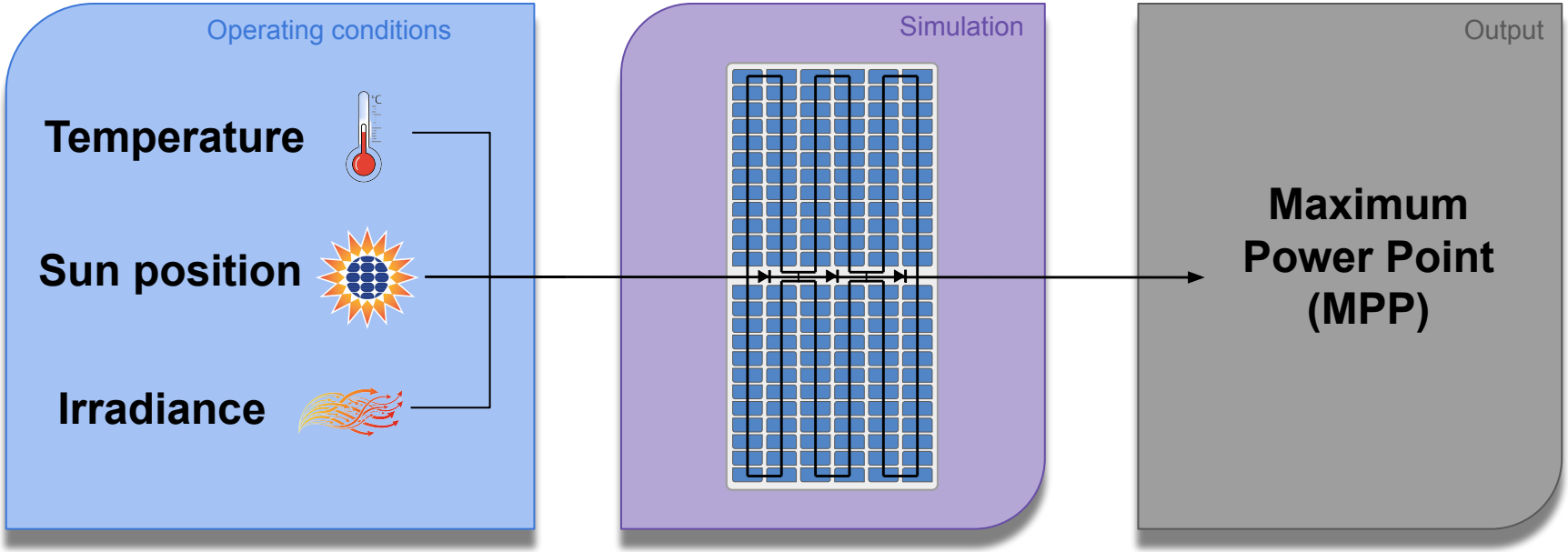
Simulation process



$(12 \times 6) \times 2 = 144$ cells

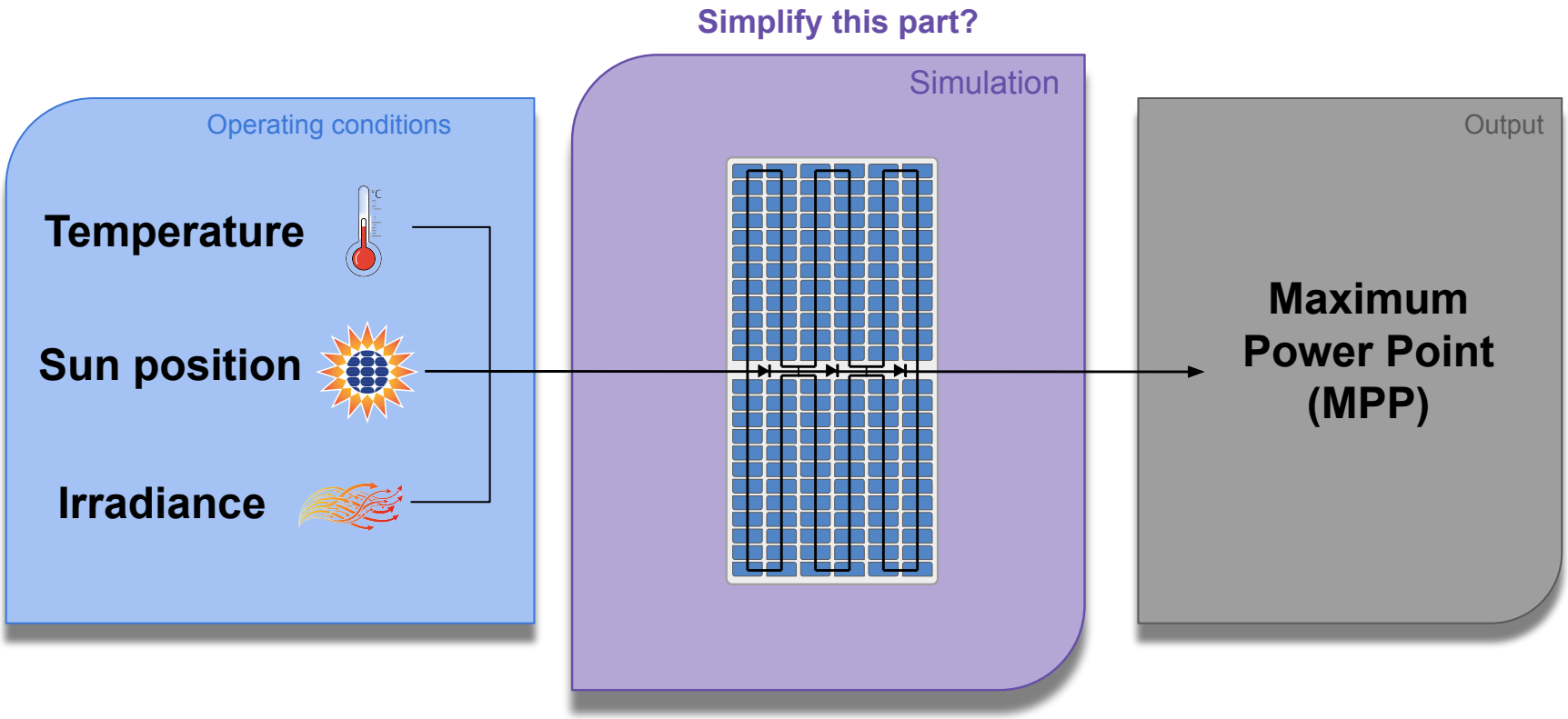
ieco.io

Simulation process



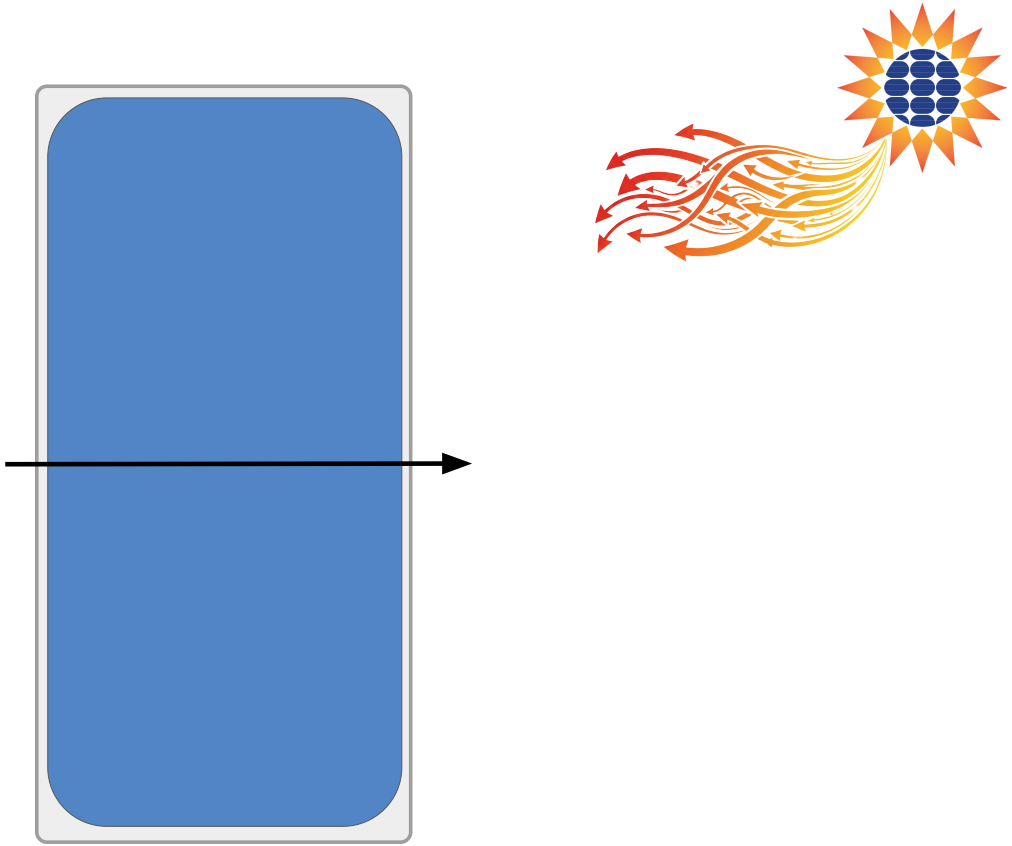
ieco.io

Simulation process



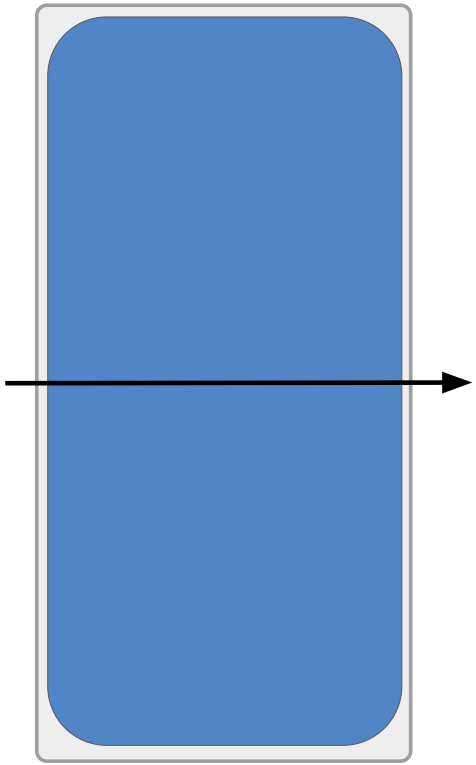
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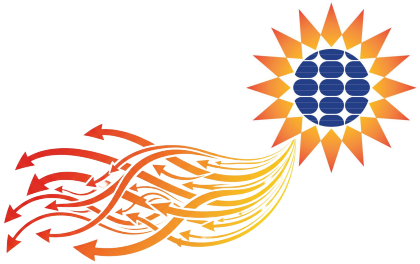
Module-level without considering shading

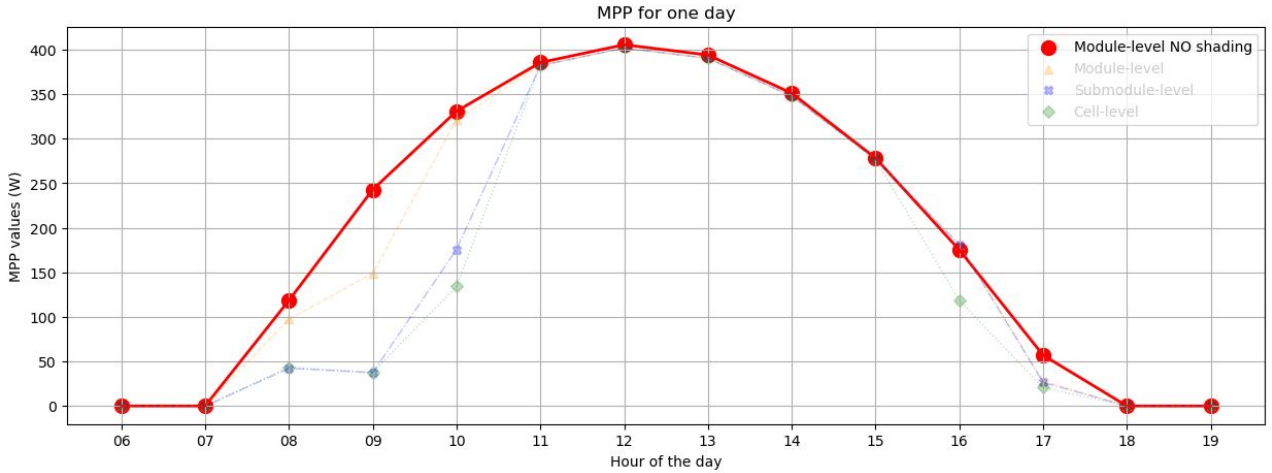


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Module-level without considering shading

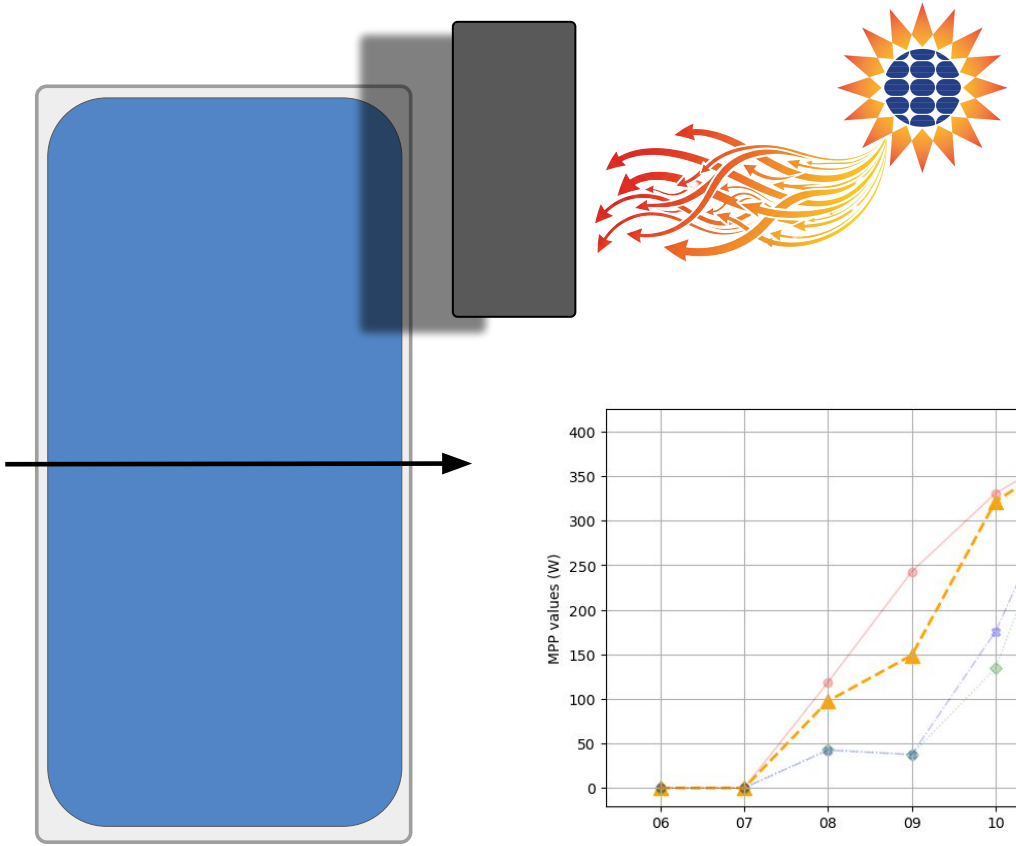


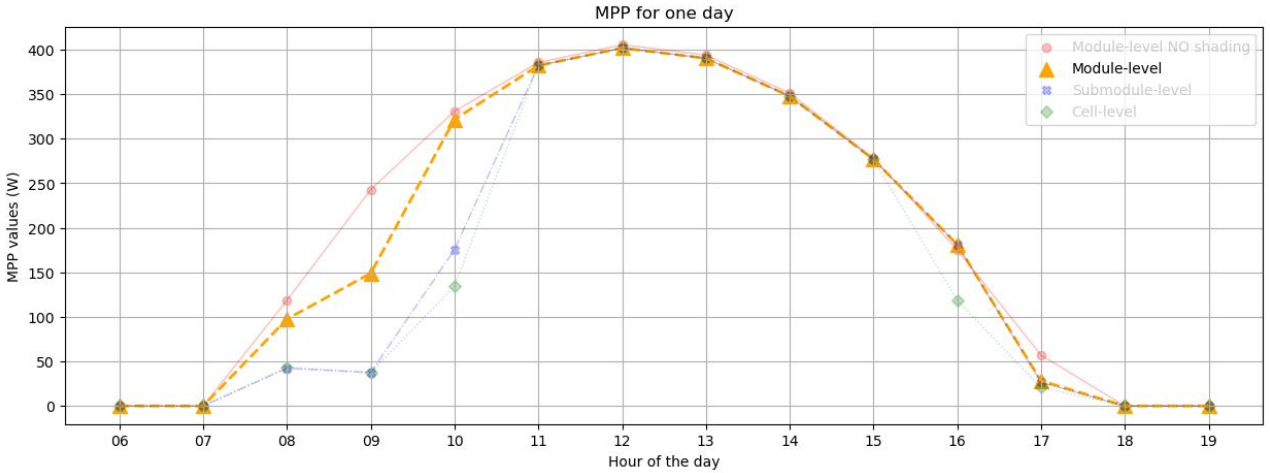




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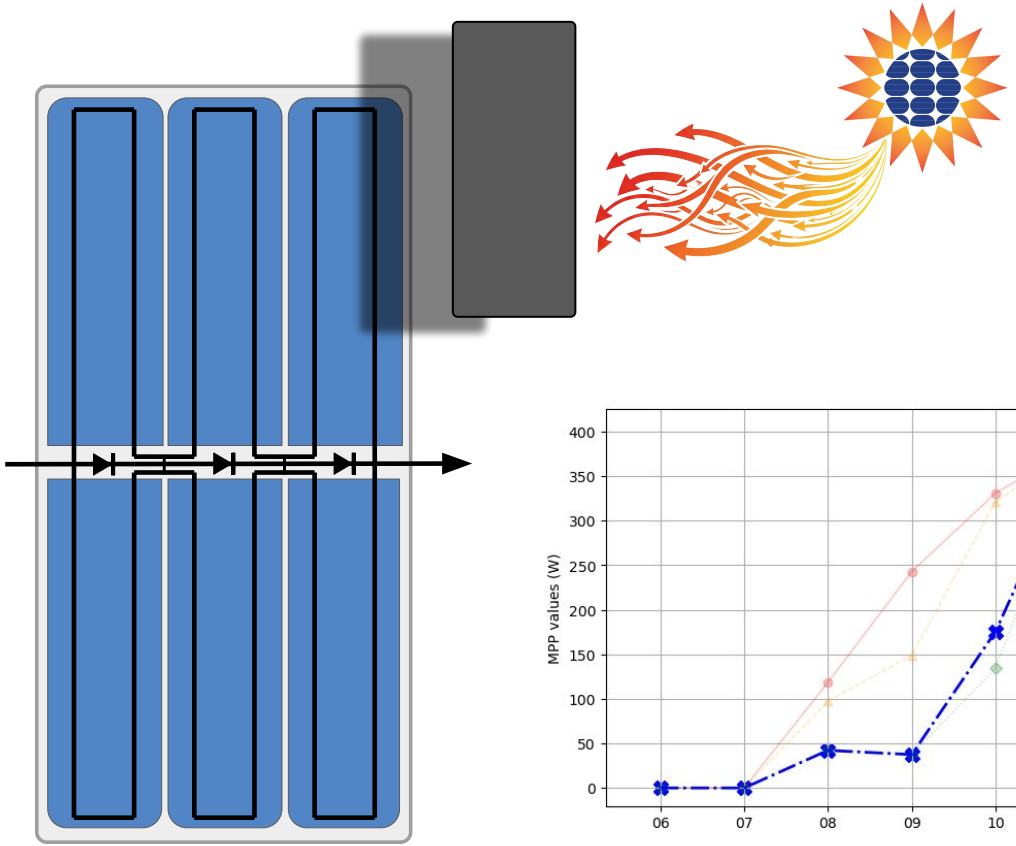
Module-level simulation

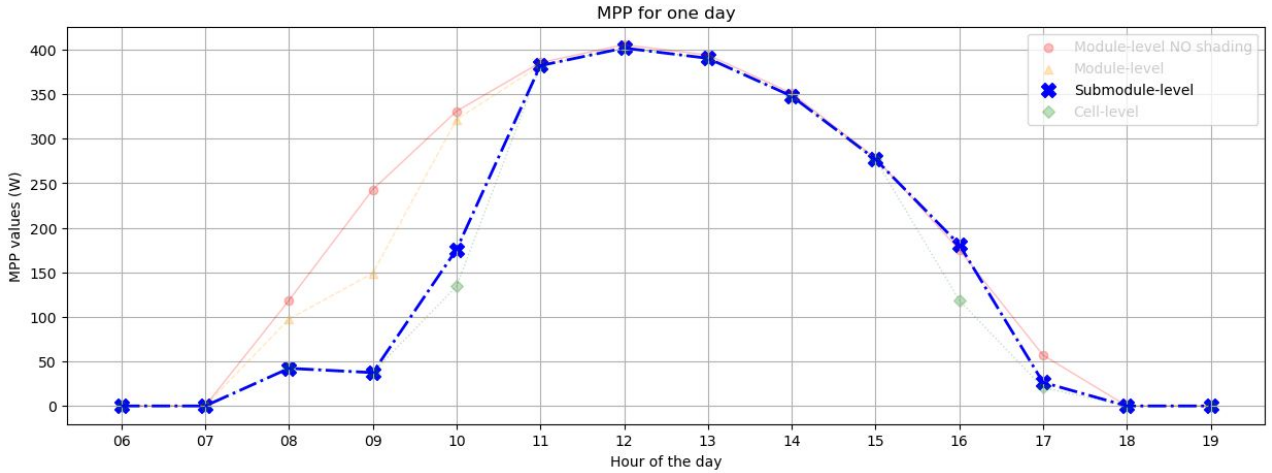




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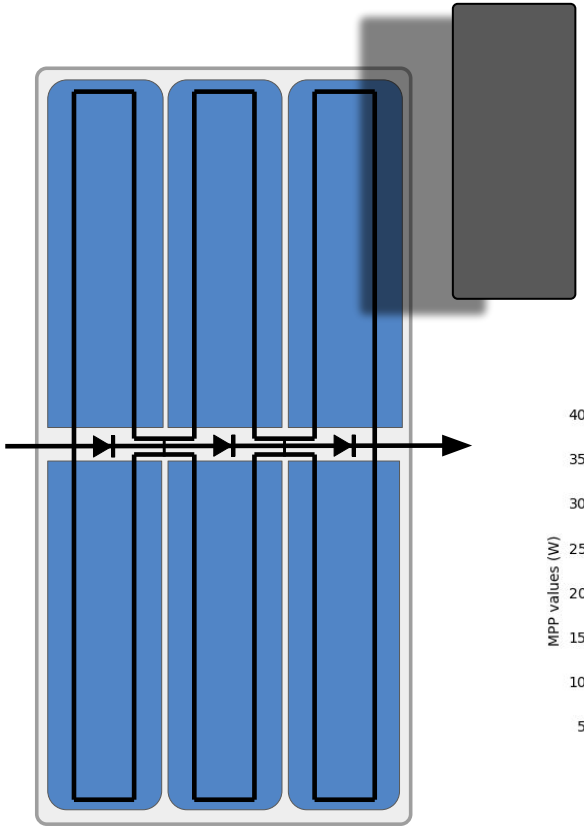
Submodule-level simulation

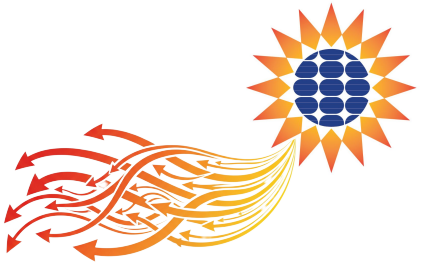




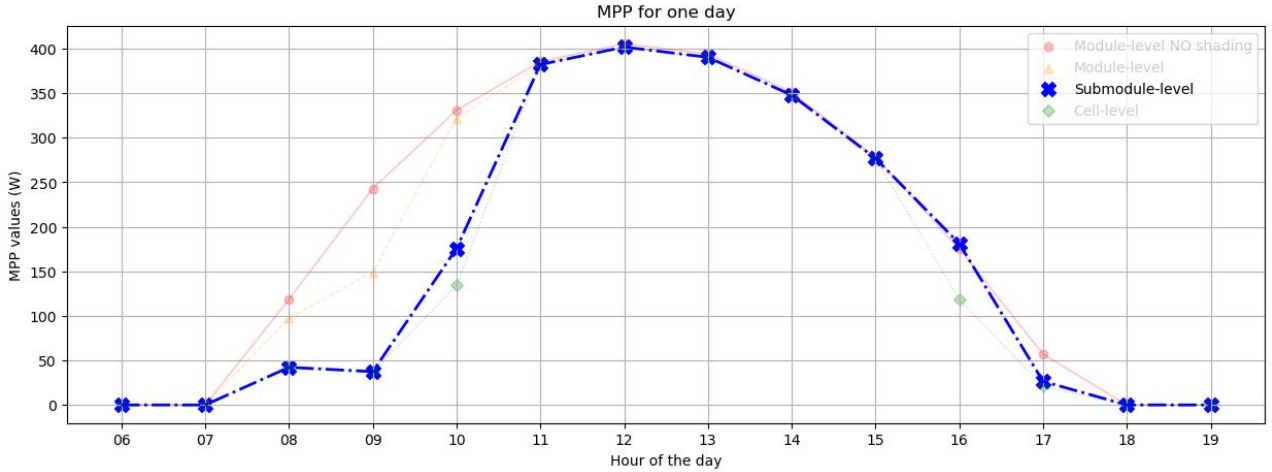
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Submodule-level simulation



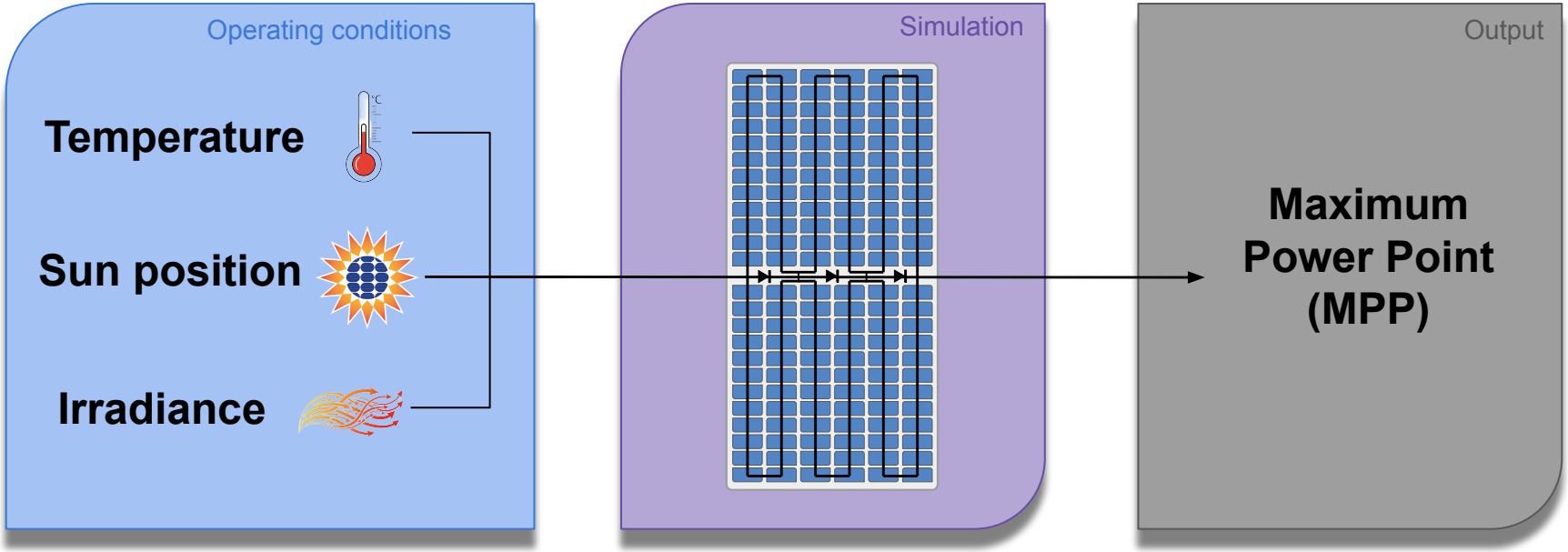


This is the one we use as the accurate reference method in our work!



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Simulation process



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Simulation process

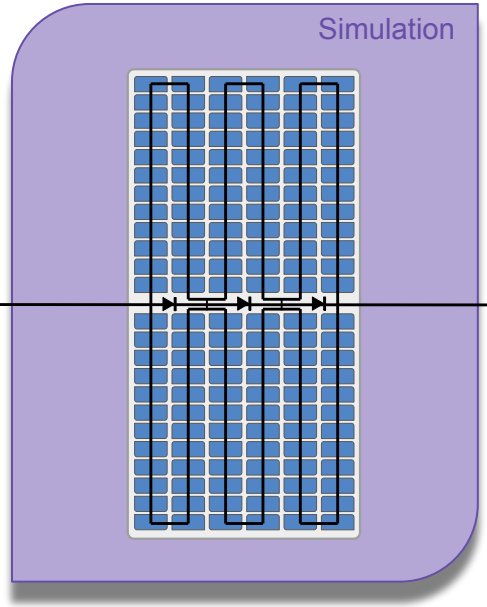
Simplify this part?

Operating conditions

Temperature

Sun position

Irradiance



Output

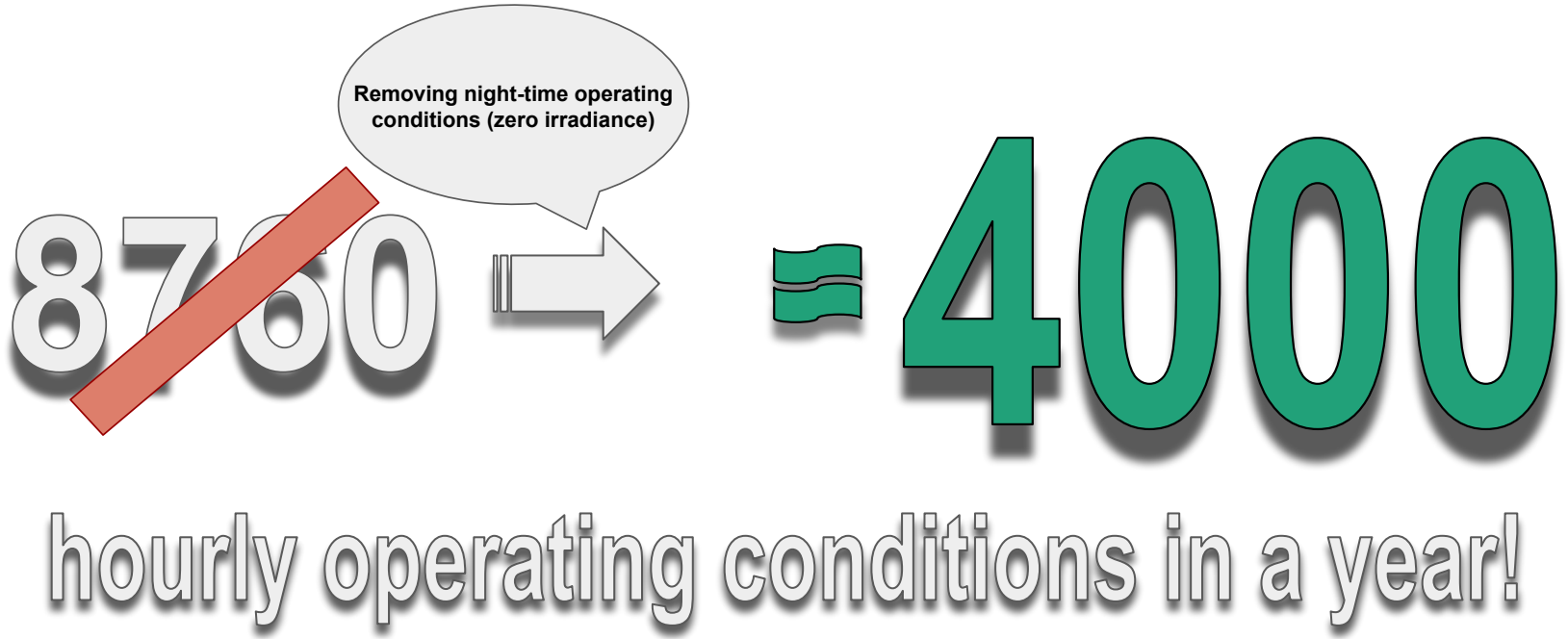
Maximum Power Point (MPP)

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8760

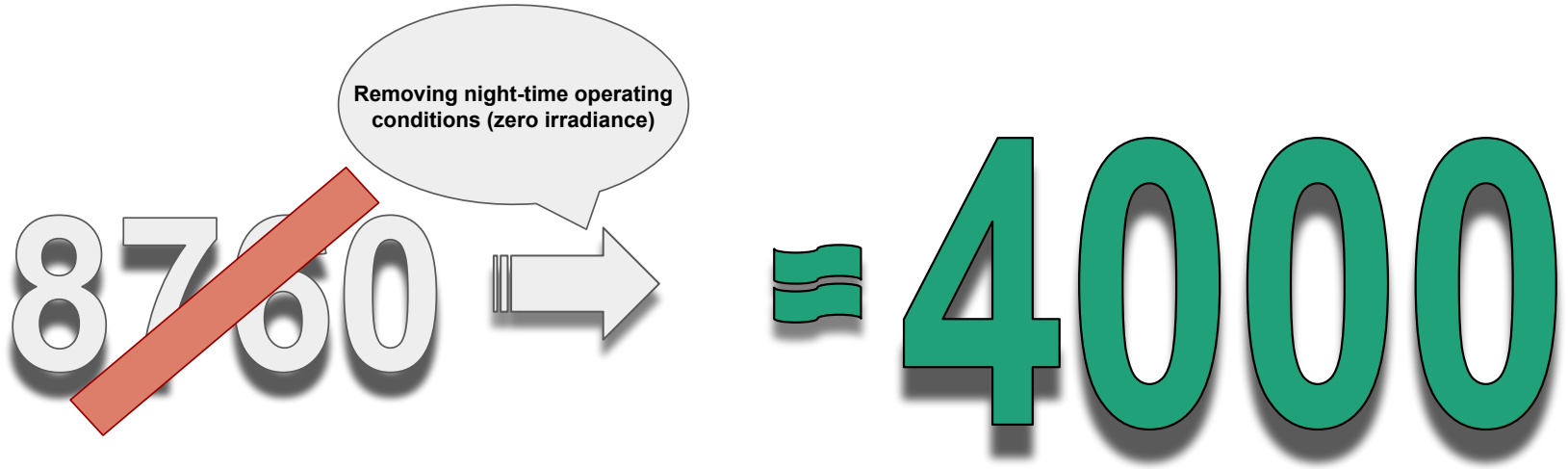
hourly operating conditions in a year!

Problem



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Problem



hourly operating conditions in a year!

high computational cost

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4000

How do we reduce this number?

Similar operating conditions produce similar outputs

2 November, 11:00



Irradiance



Sun elevation

Sun azimuth



Temperature

3 November, 11:00



Irradiance



Sun elevation

Sun azimuth



Temperature

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Similar operating conditions produce similar outputs

2 November, 11:00

Irradiance	922.49 W/m²
Sun elevation	31.14°
Sun azimuth	-24.87°
Temperature	43.19°C

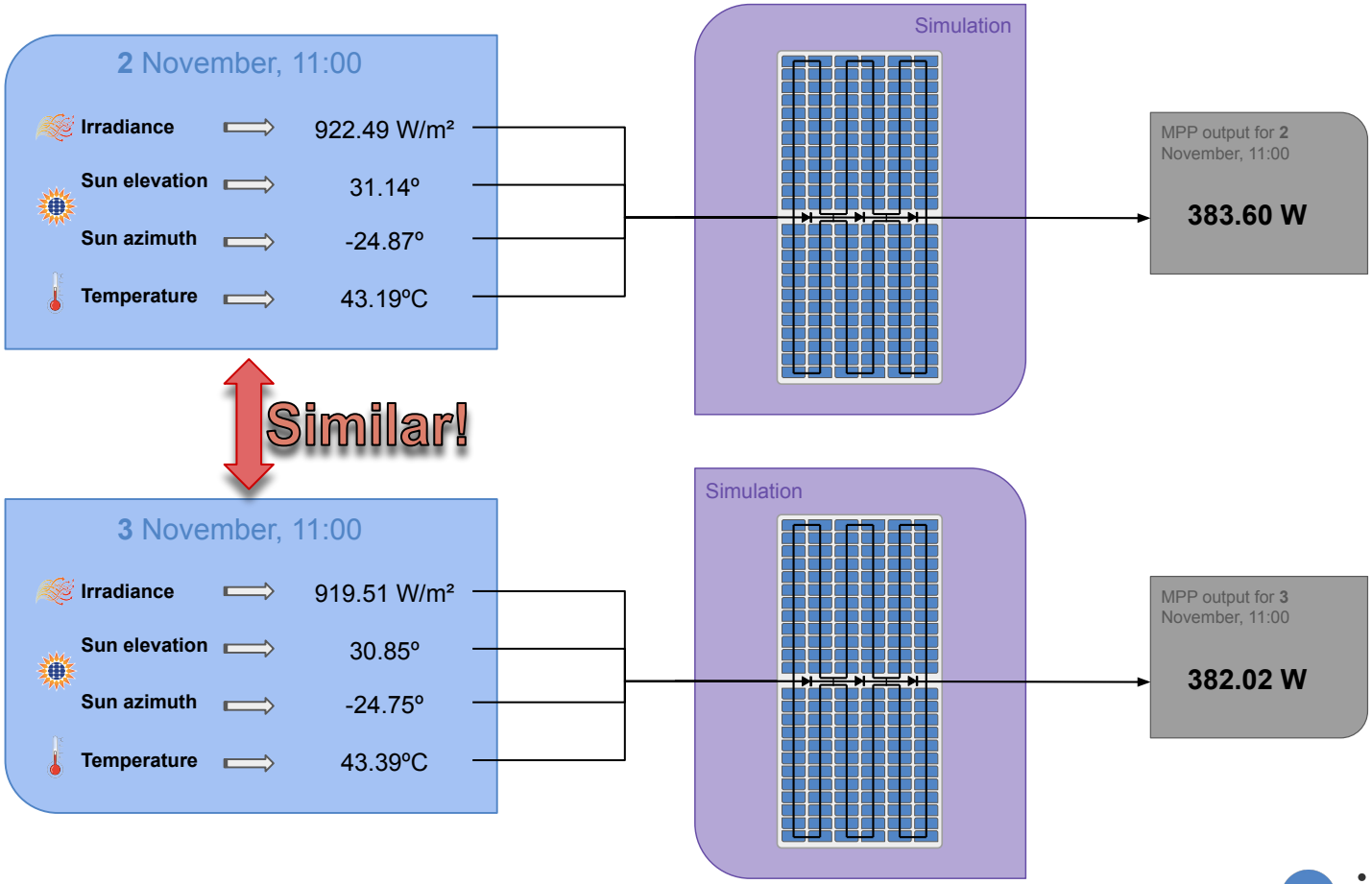
Similar!

3 November, 11:00

Irradiance	919.51 W/m²
Sun elevation	30.85°
Sun azimuth	-24.75°
Temperature	43.39°C

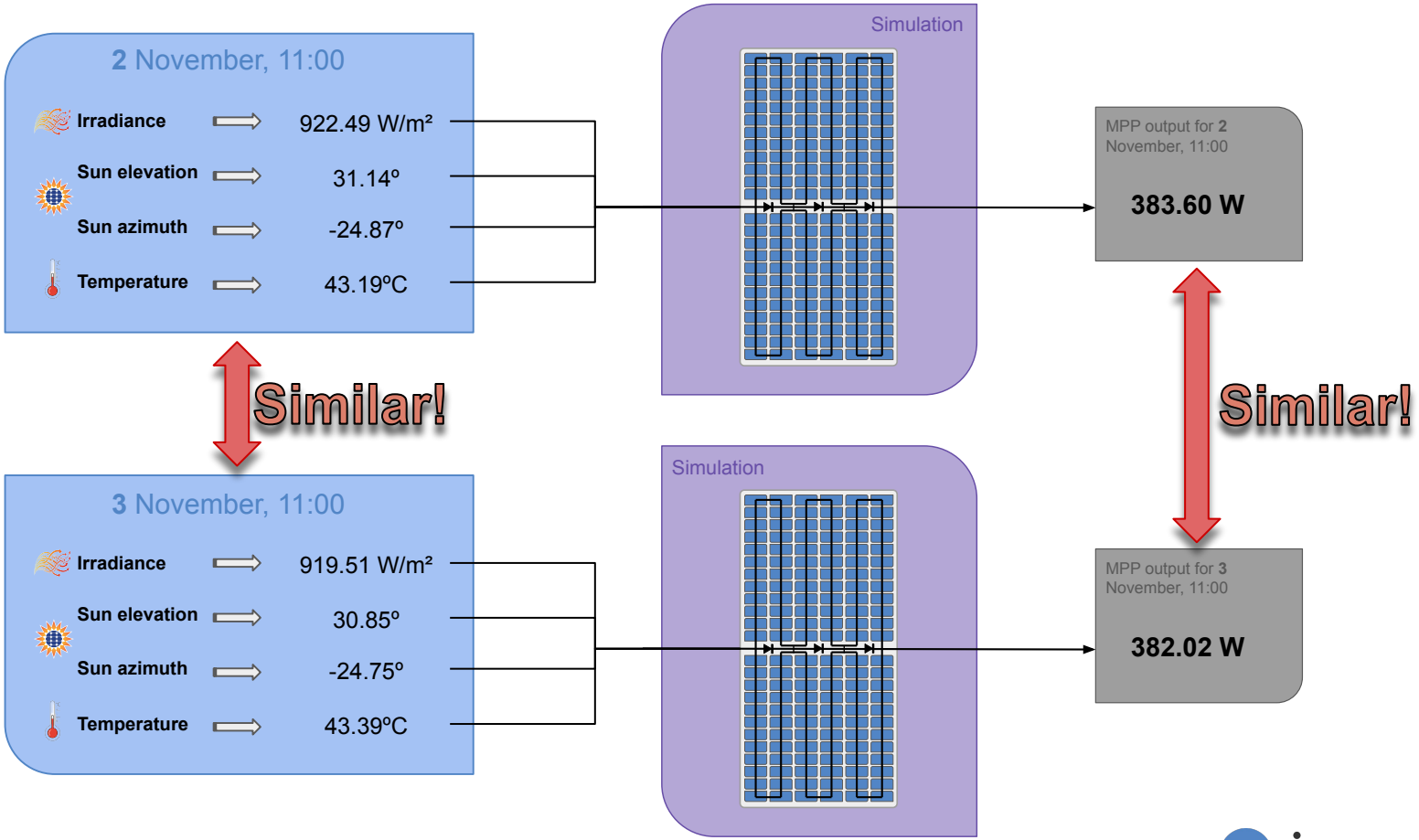
ieco.io

Similar operating conditions produce similar outputs



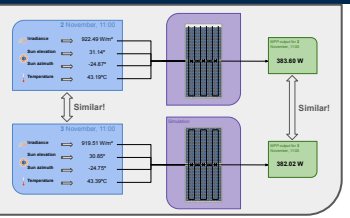
ieco.io

Similar operating conditions produce similar outputs



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Aggregation of similar operating conditions



2 November, 11:00

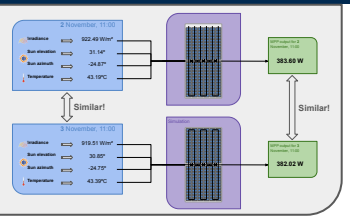
Irradiance	⟹	922.49 W/m²
Sun elevation	⟹	31.14°
Sun azimuth	⟹	-24.87°
Temperature	⟹	43.19°C

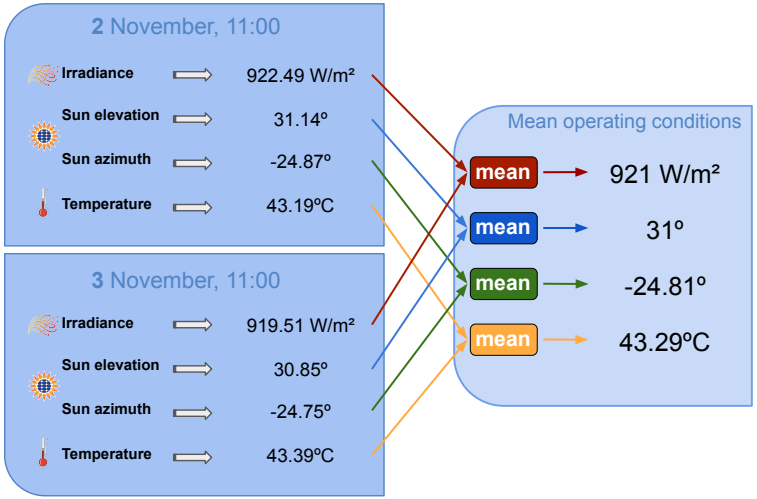
3 November, 11:00

Irradiance	⟹	919.51 W/m²
Sun elevation	⟹	30.85°
Sun azimuth	⟹	-24.75°
Temperature	⟹	43.39°C

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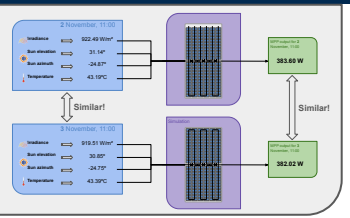
Aggregation of similar operating conditions

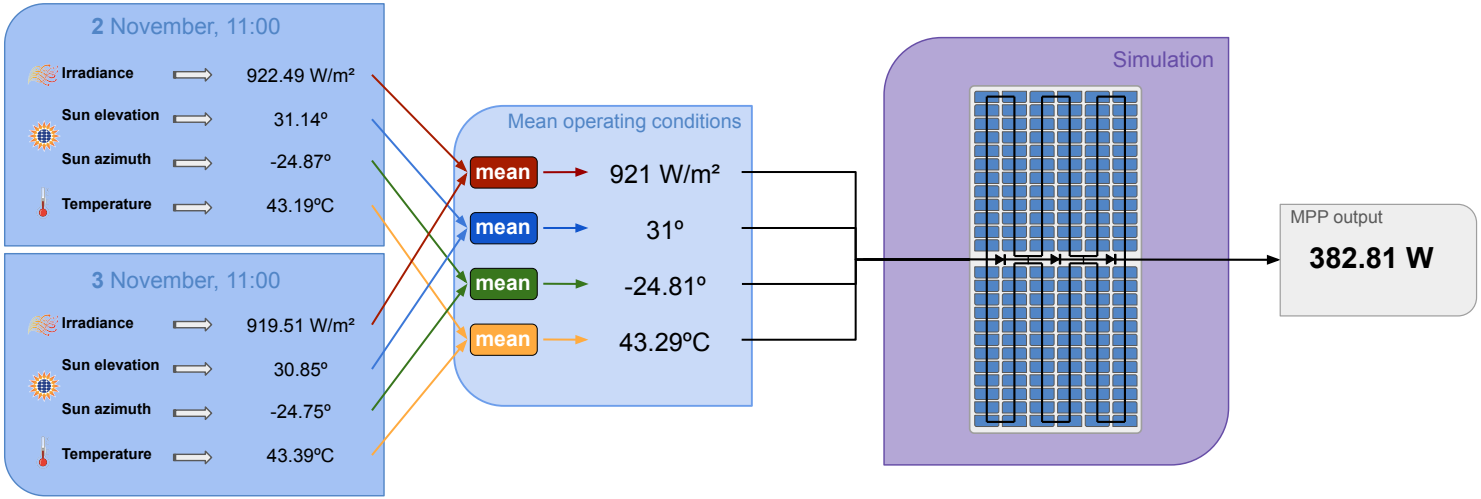




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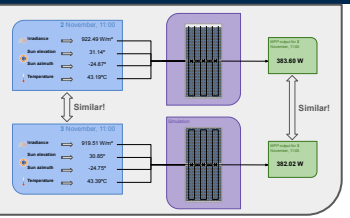
Aggregation of similar operating conditions





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Aggregation of similar operating conditions



2 November, 11:00

Irradiance	922.49 W/m²
Sun elevation	31.14°
Sun azimuth	-24.87°
Temperature	43.19°C

3 November, 11:00

Irradiance	919.51 W/m²
Sun elevation	30.85°
Sun azimuth	-24.75°
Temperature	43.39°C

Mean operating conditions

mean	921 W/m²
mean	31°
mean	-24.81°
mean	43.29°C

Simulation

MPP output

382.81 W

MPP output for **2** November, 11:00

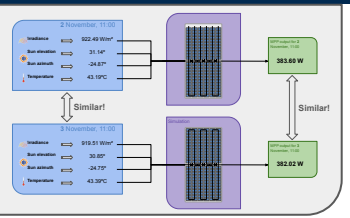
382.81 W

MPP output for **3** November, 11:00

382.81 W

ieco.io

Aggregation of similar operating conditions



2 November, 11:00

- Irradiance → 922.49 W/m²
- Sun elevation → 31.14°
- Sun azimuth → -24.87°
- Temperature → 43.19°C

3 November, 11:00

- Irradiance → 919.51 W/m²
- Sun elevation → 30.85°
- Sun azimuth → -24.75°
- Temperature → 43.39°C

Mean operating conditions

- mean → 921 W/m²
- mean → 31°
- mean → -24.81°
- mean → 43.29°C

Simulation

MPP output

382.81 W

Real value: 383.60 W

MPP output for **2** November, 11:00

382.81 W

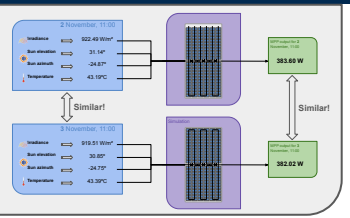
MPP output for **3** November, 11:00

382.81 W

Real value: 382.02 W

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Aggregation of similar operating conditions



Just one simulation!

2 November, 11:00

Irradiance	922.49 W/m²
Sun elevation	31.14°
Sun azimuth	-24.87°
Temperature	43.19°C

3 November, 11:00

Irradiance	919.51 W/m²
Sun elevation	30.85°
Sun azimuth	-24.75°
Temperature	43.39°C

Mean operating conditions

mean	921 W/m²
mean	31°
mean	-24.81°
mean	43.29°C

Simulation

MPP output

382.81 W

Real value:
383.60 W

MPP output for **2** November, 11:00

382.81 W

MPP output for **3** November, 11:00

382.81 W

Real value:
382.02 W

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How do we automatically aggregate similar operating conditions?

Aggregation of similar operating conditions

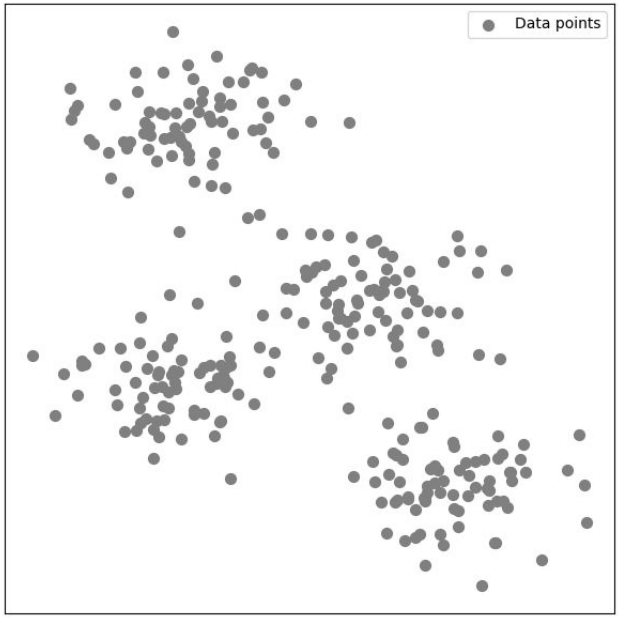
How do we automatically aggregate similar operating conditions?

k-means

Unsupervised machine learning algorithm for data segmentation (clustering)

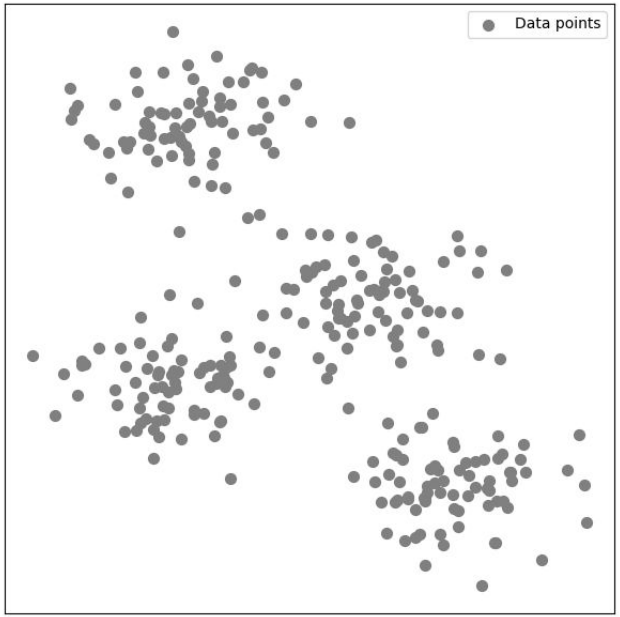
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K-means algorithm

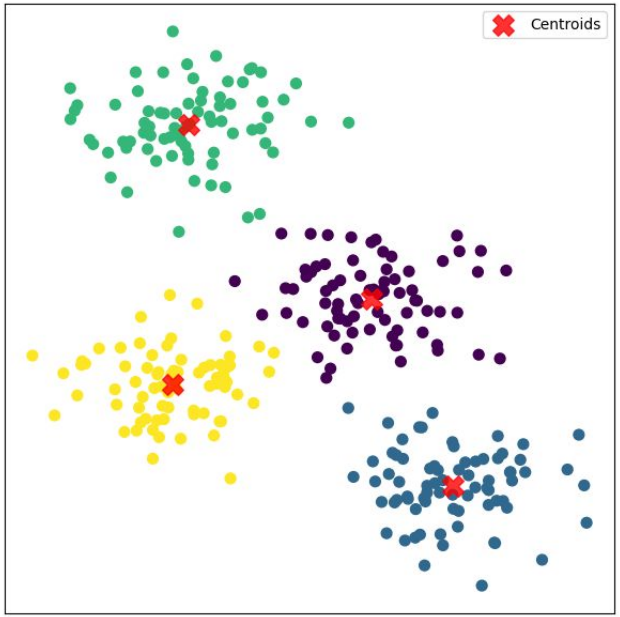


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K-means algorithm

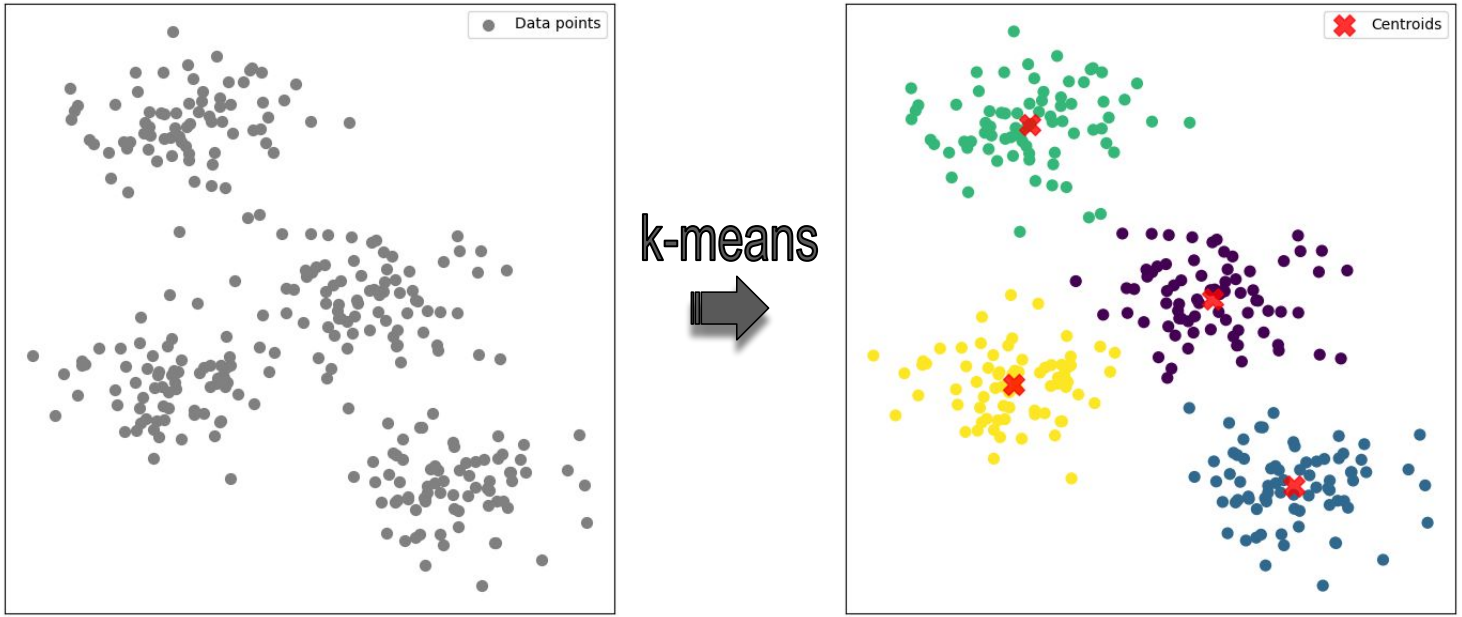


k-means



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K-means algorithm



Centroids (✖) as new data points

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Initial data aggregation approaches

- StraightForward Aggregation (SFA)

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- StraightForward Aggregation (SFA)
- Hierarchical Hourly Aggregation (HHA)

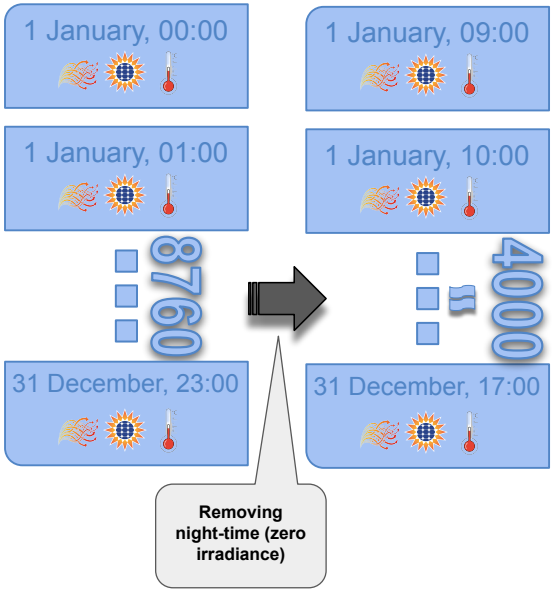
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Initial data aggregation approaches

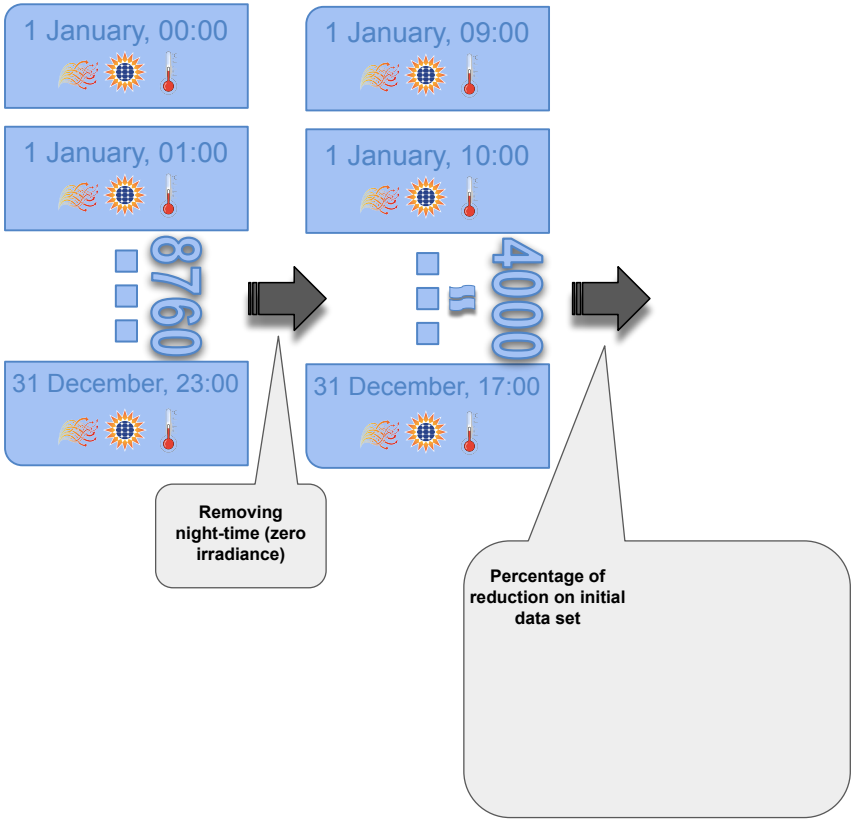
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Straight-Forward Aggregation (SFA)

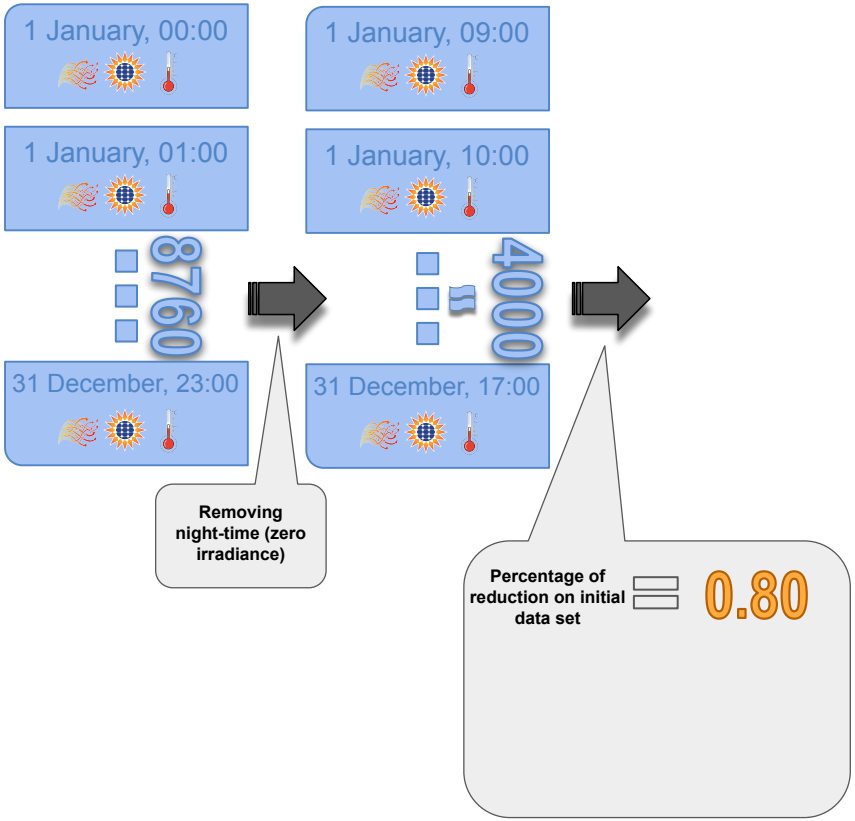


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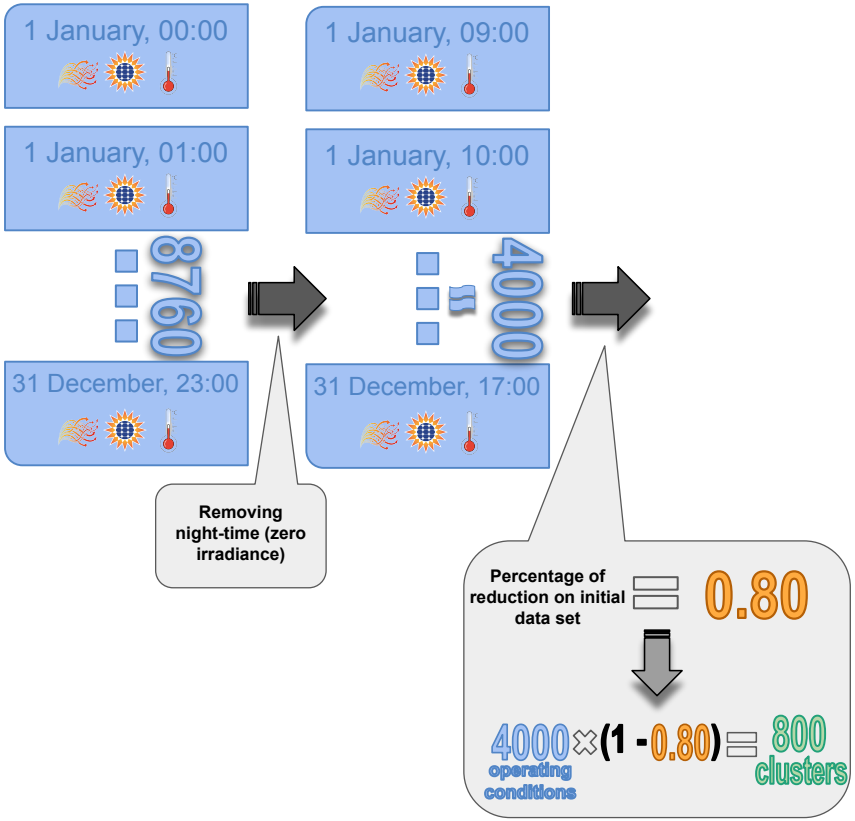
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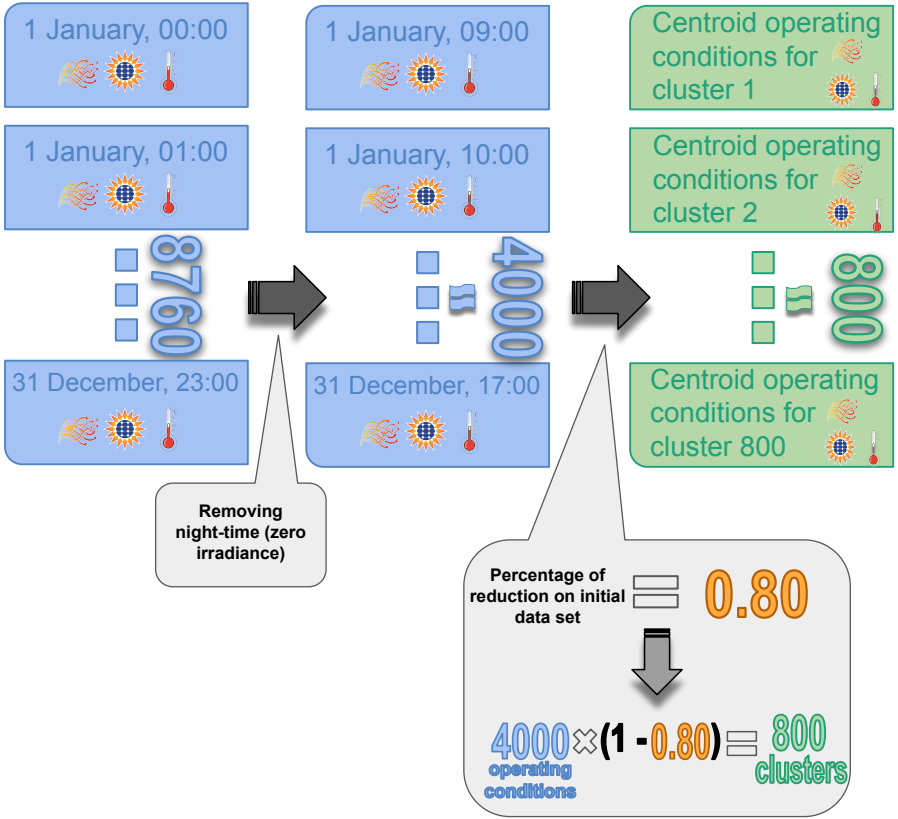
ieco.io

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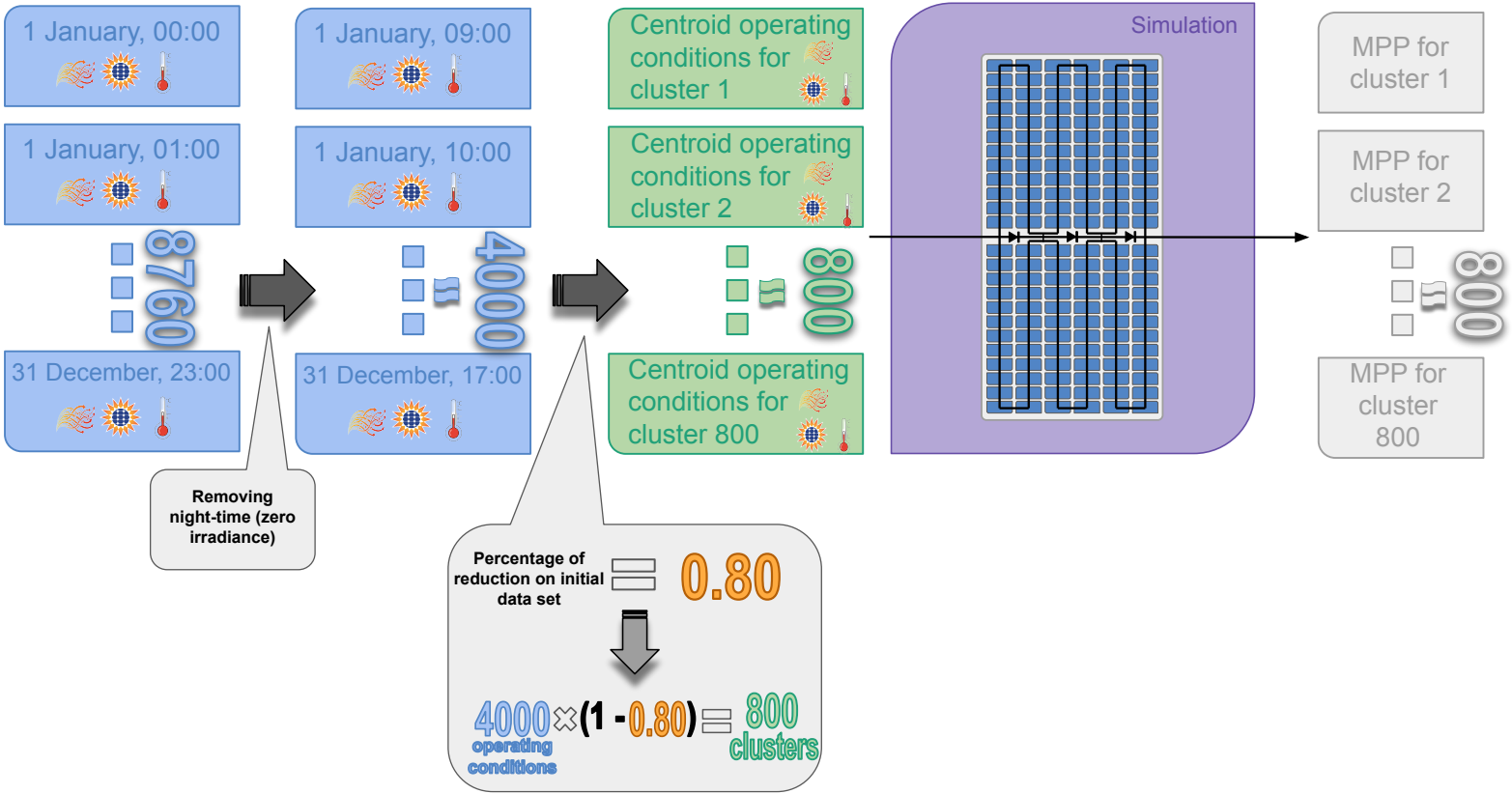
ieco.io

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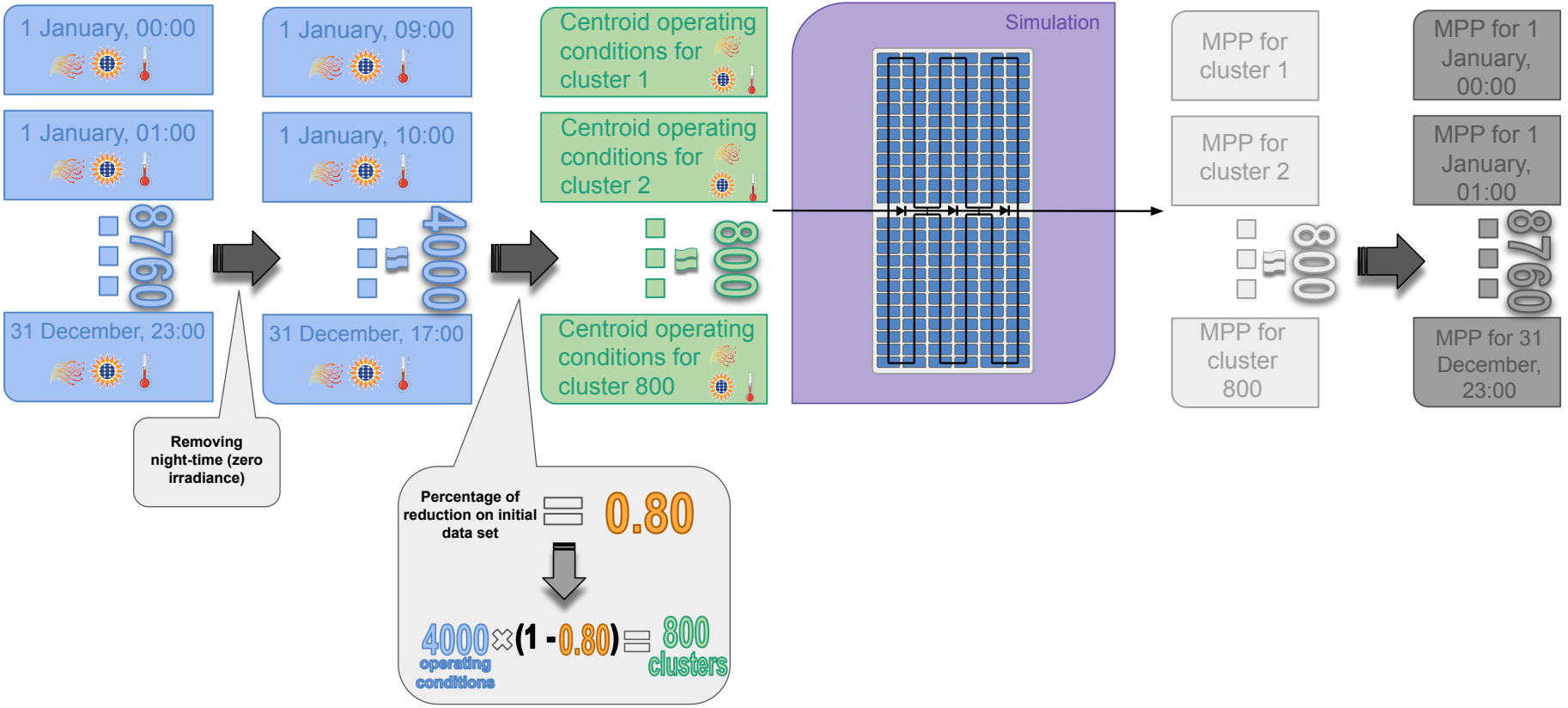
ieco.io

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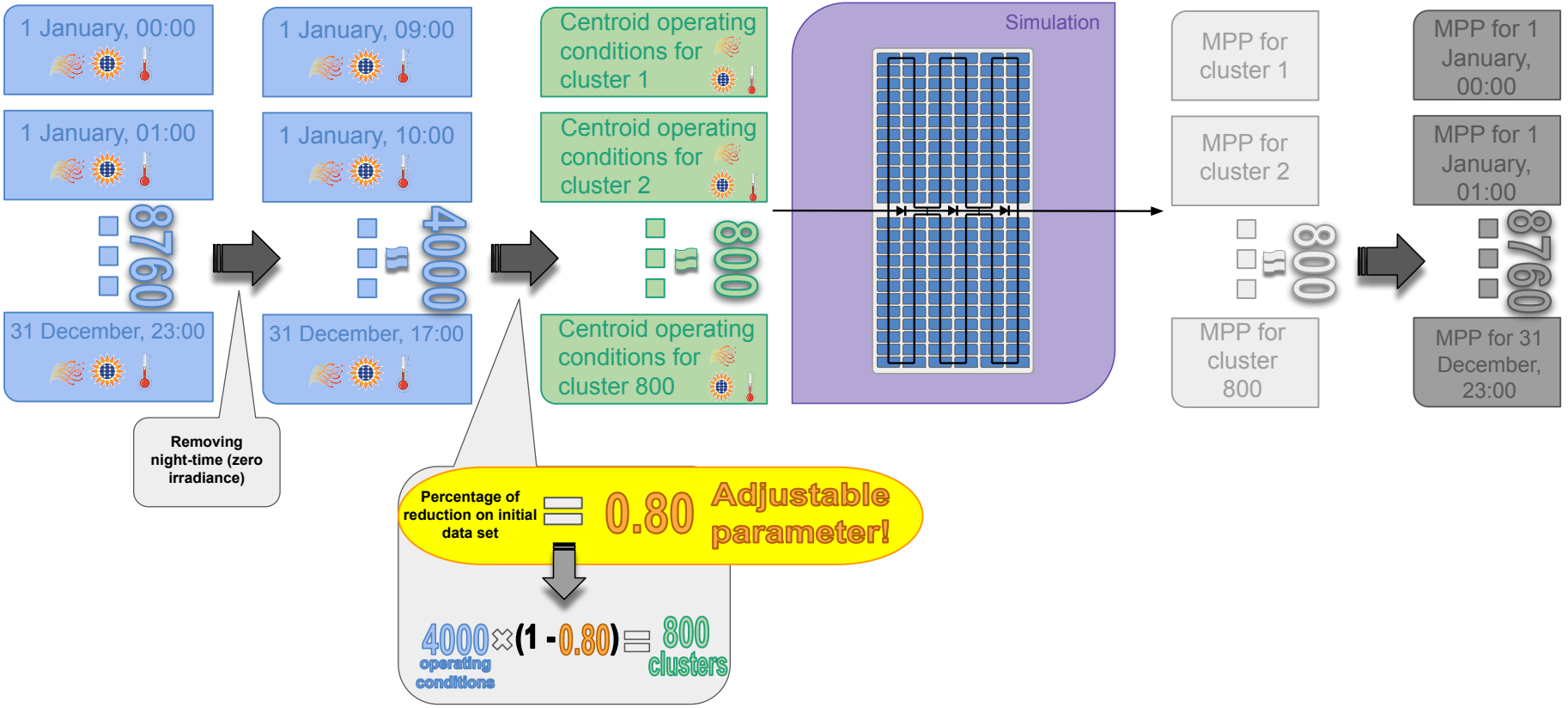
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Straight-Forward Aggregation (SFA)



ieco.io

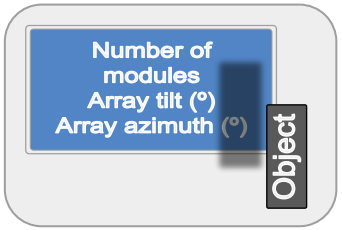
Straight-Forward Aggregation (SFA)



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Simulation scenarios for performance testing

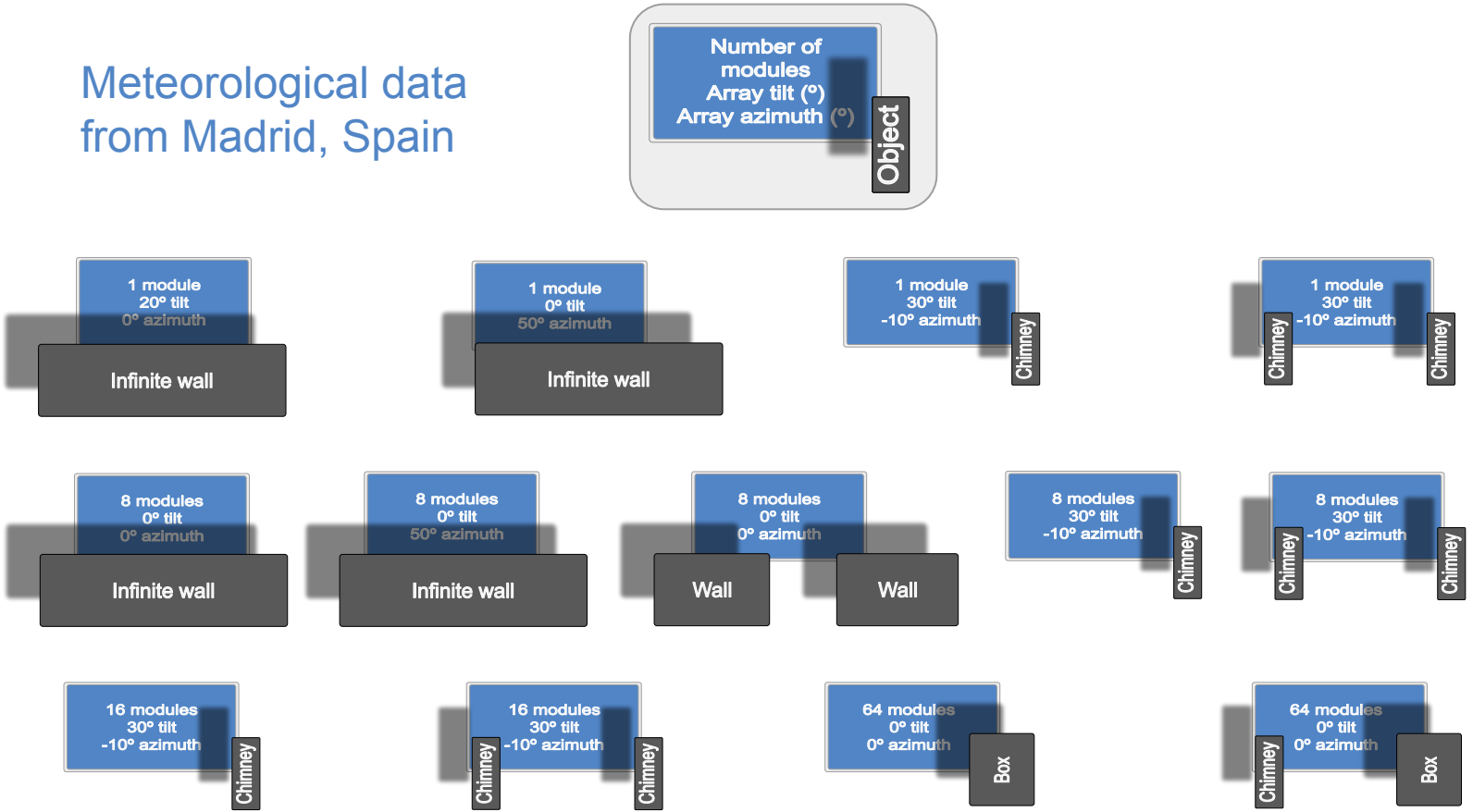
Meteorological data
from Madrid, Spain



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Simulation scenarios for performance testing

Meteorological data
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SFA performance results

$$Relative\ Error = \frac{\sum \left| MPP_{new\ method} - MPP_{reference\ method} \right|}{\sum MPP_{reference\ method}}$$

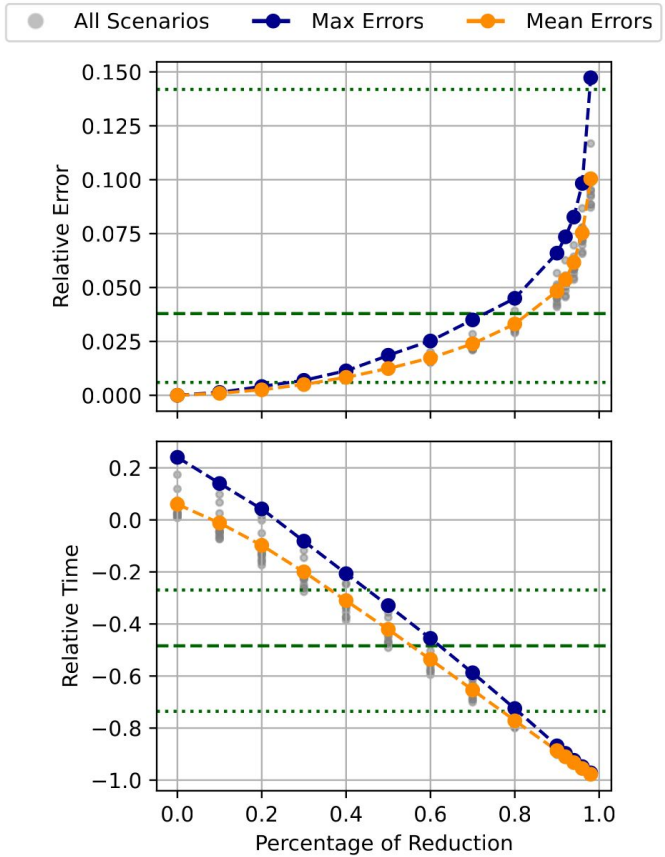
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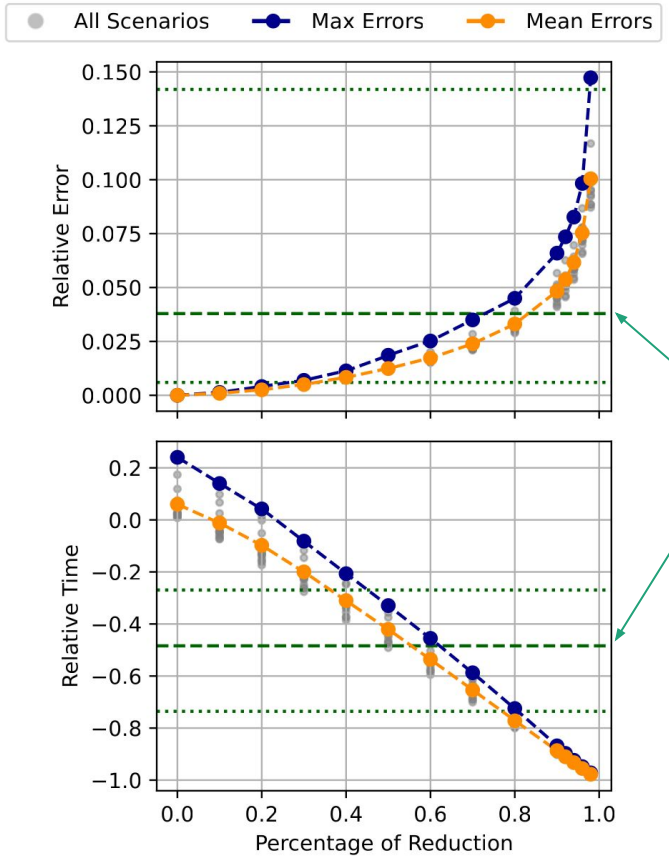


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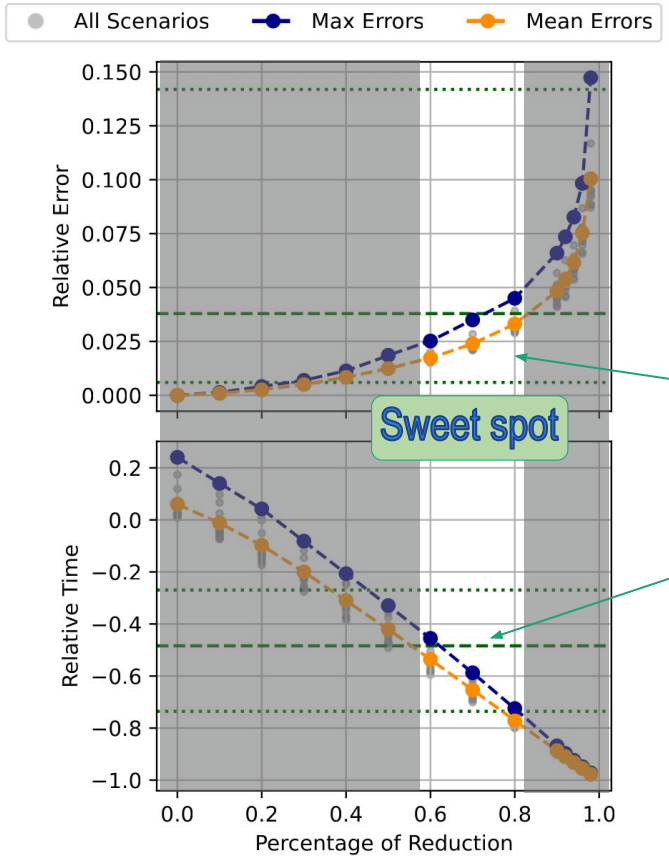
Module-level performance for comparison

ieco.io

SFA performance results

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Less errors than module-level simulation

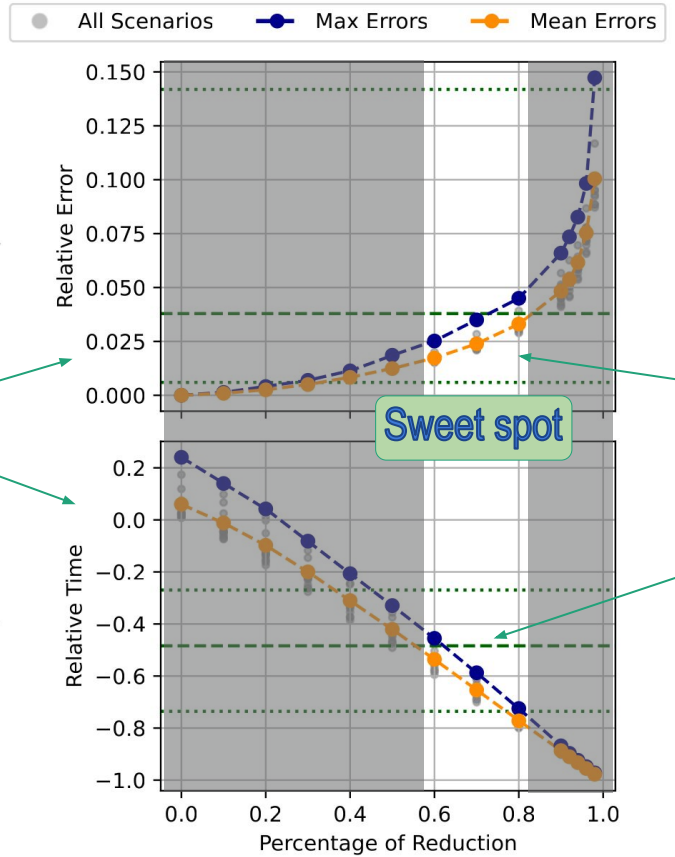
More reduction in computational time than module-level simulation

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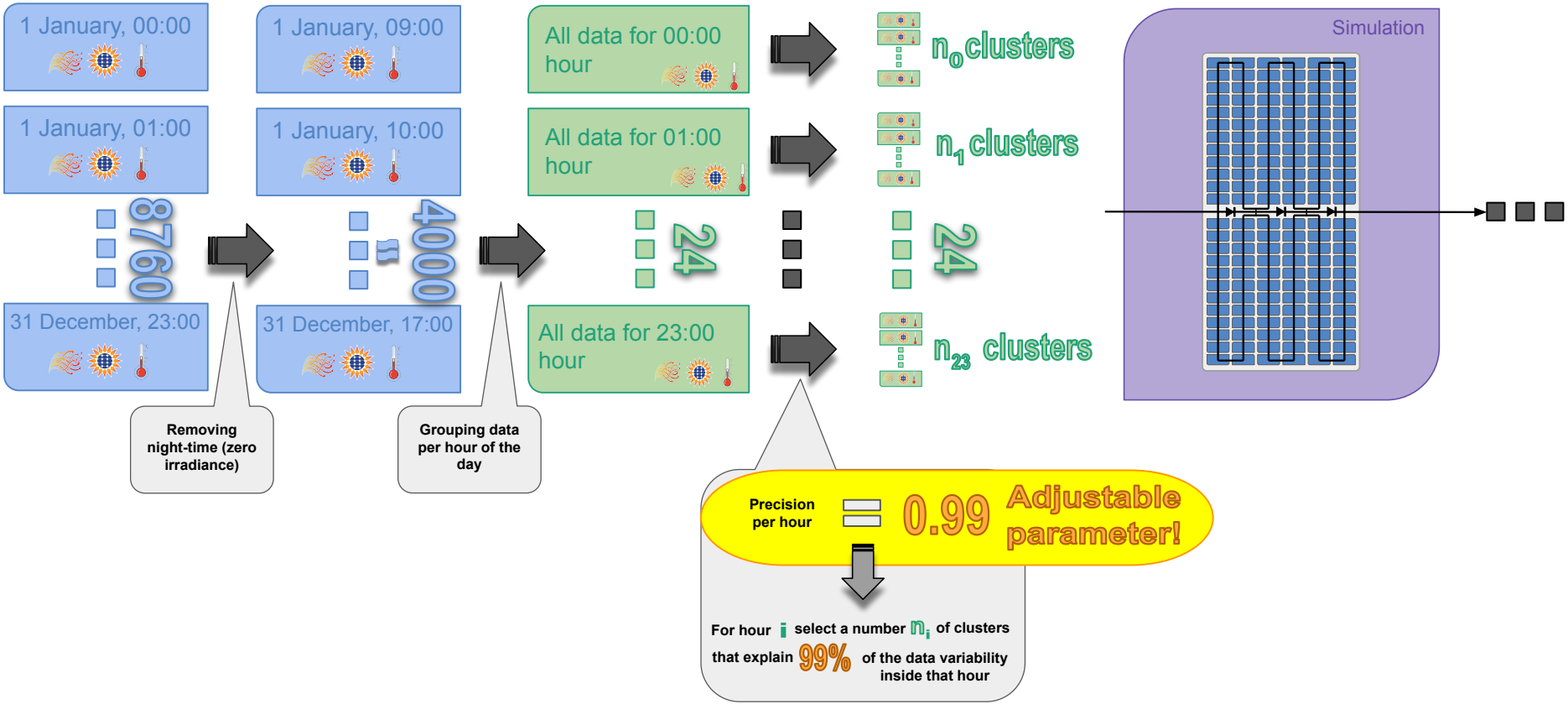
More stable results!

Less errors than module-level simulation

More reduction in computational time than module-level simulation

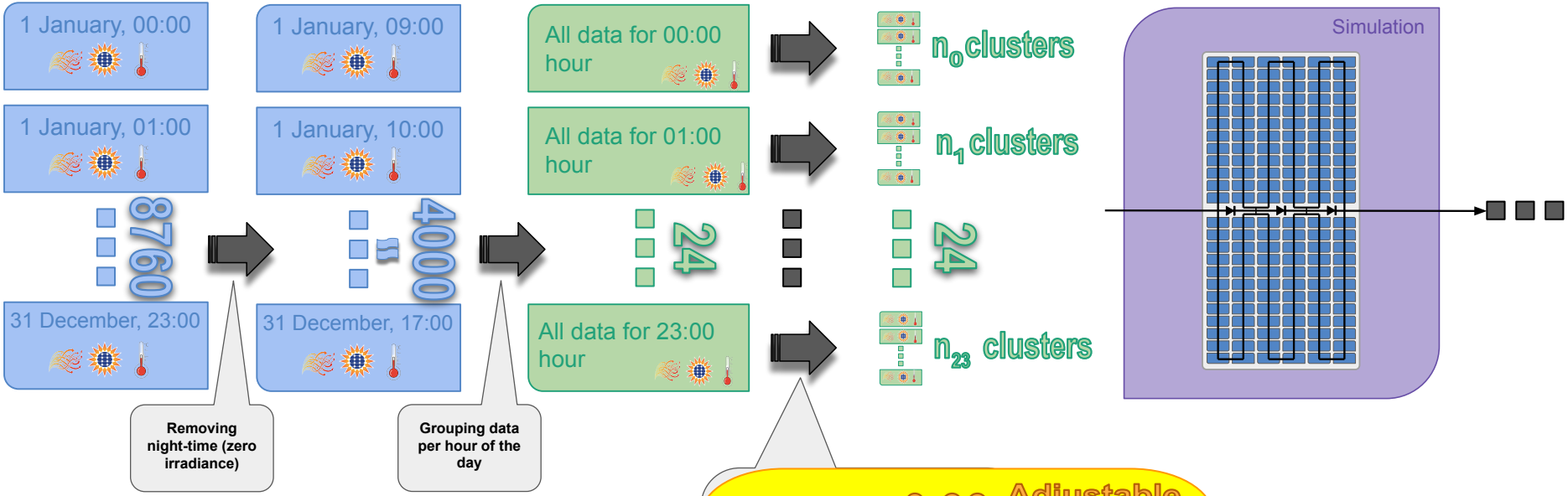
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Hierarchical Hourly Aggregation (HHA)



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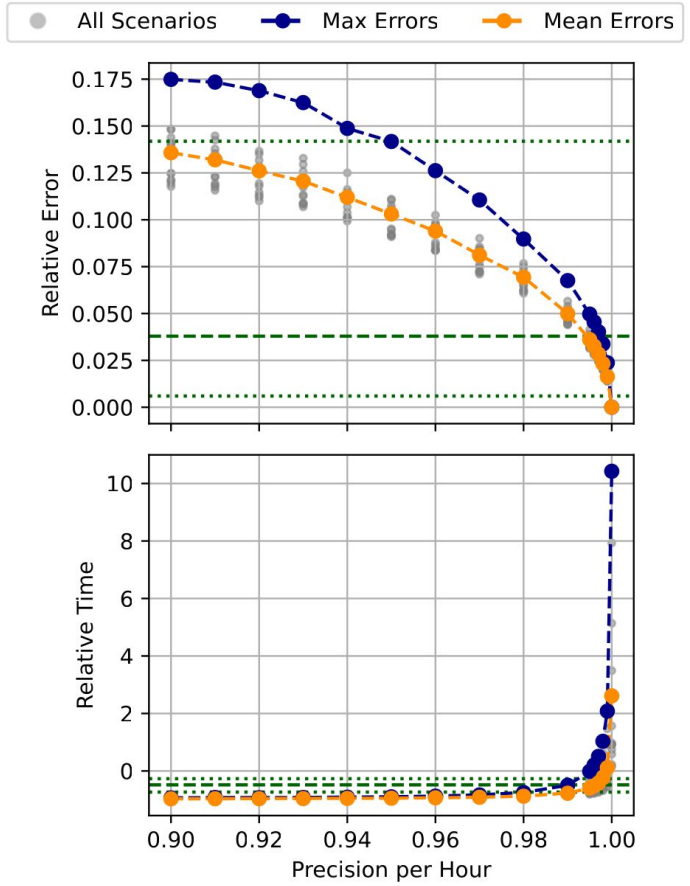
Hierarchical Hourly Aggregation (HHA)



more complex and less interpretable way of selecting the number of clusters compared with SFA

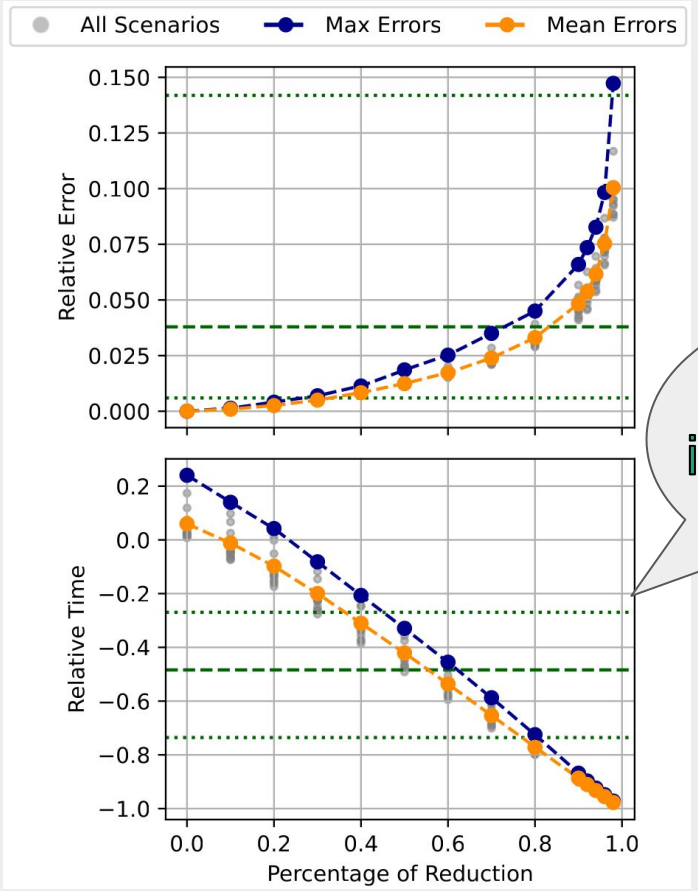
ieco.io

HHA performance



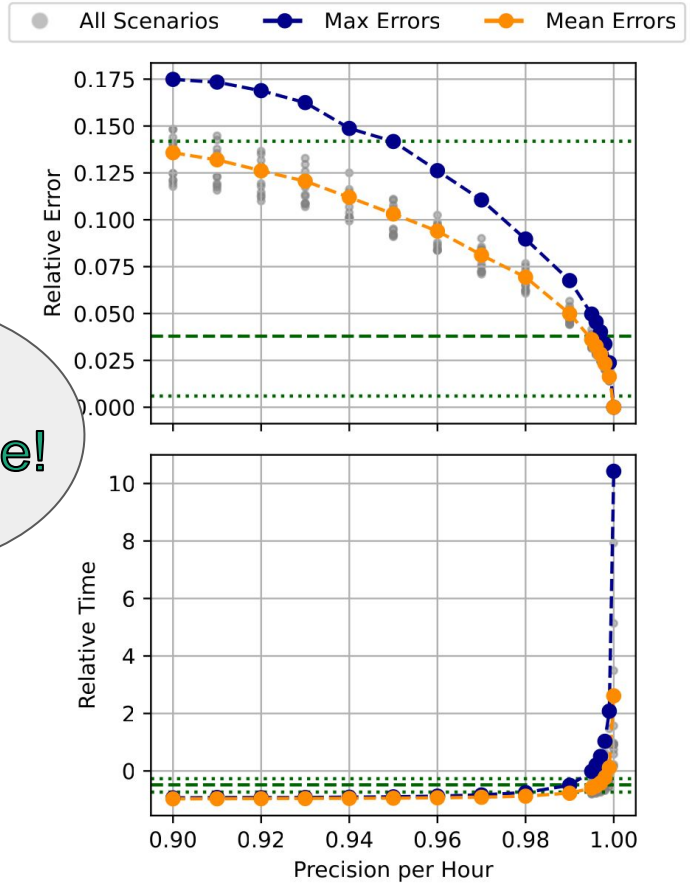
ieco.io

SFA performance



More interpretable!

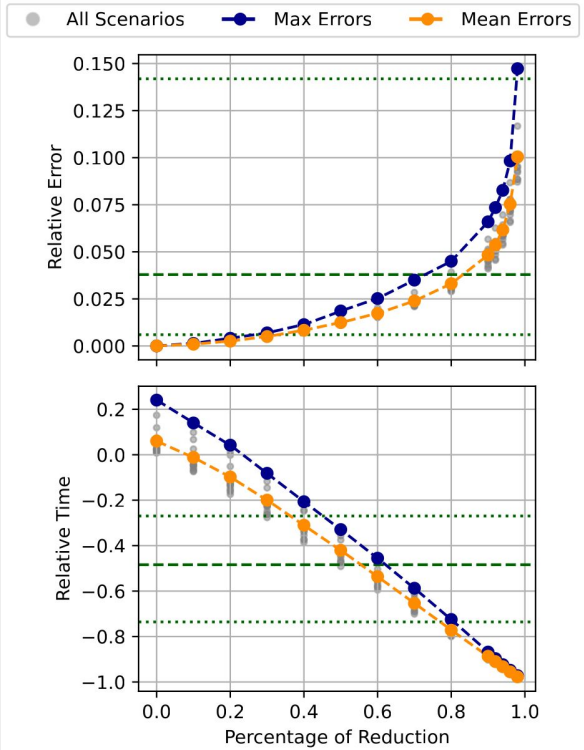
HHA performance



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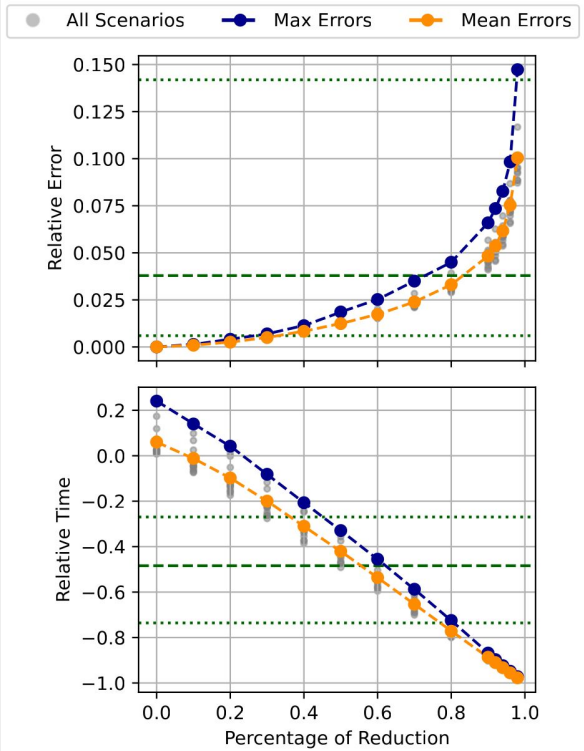
How do we compare both methods?

How do we compare both methods?



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How do we compare both methods?



mean

$$Relative\ Error = \frac{\sum \left| MPP_{new\ method} - MPP_{reference\ method} \right|}{\sum MPP_{reference\ method}}$$

VS

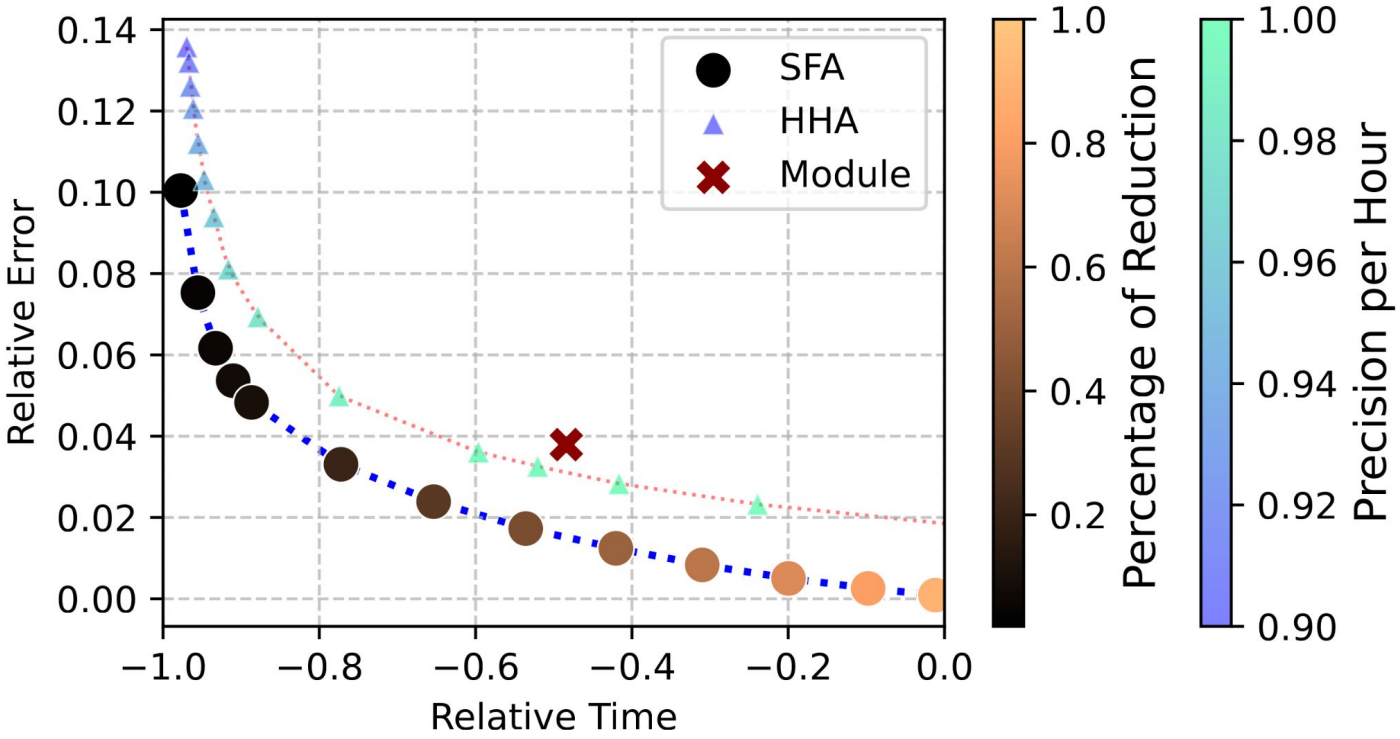
mean

$$Relative\ Time = \frac{time_{new\ method} - time_{reference\ method}}{time_{reference\ method}}$$

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Methods comparison (Pareto Frontier)

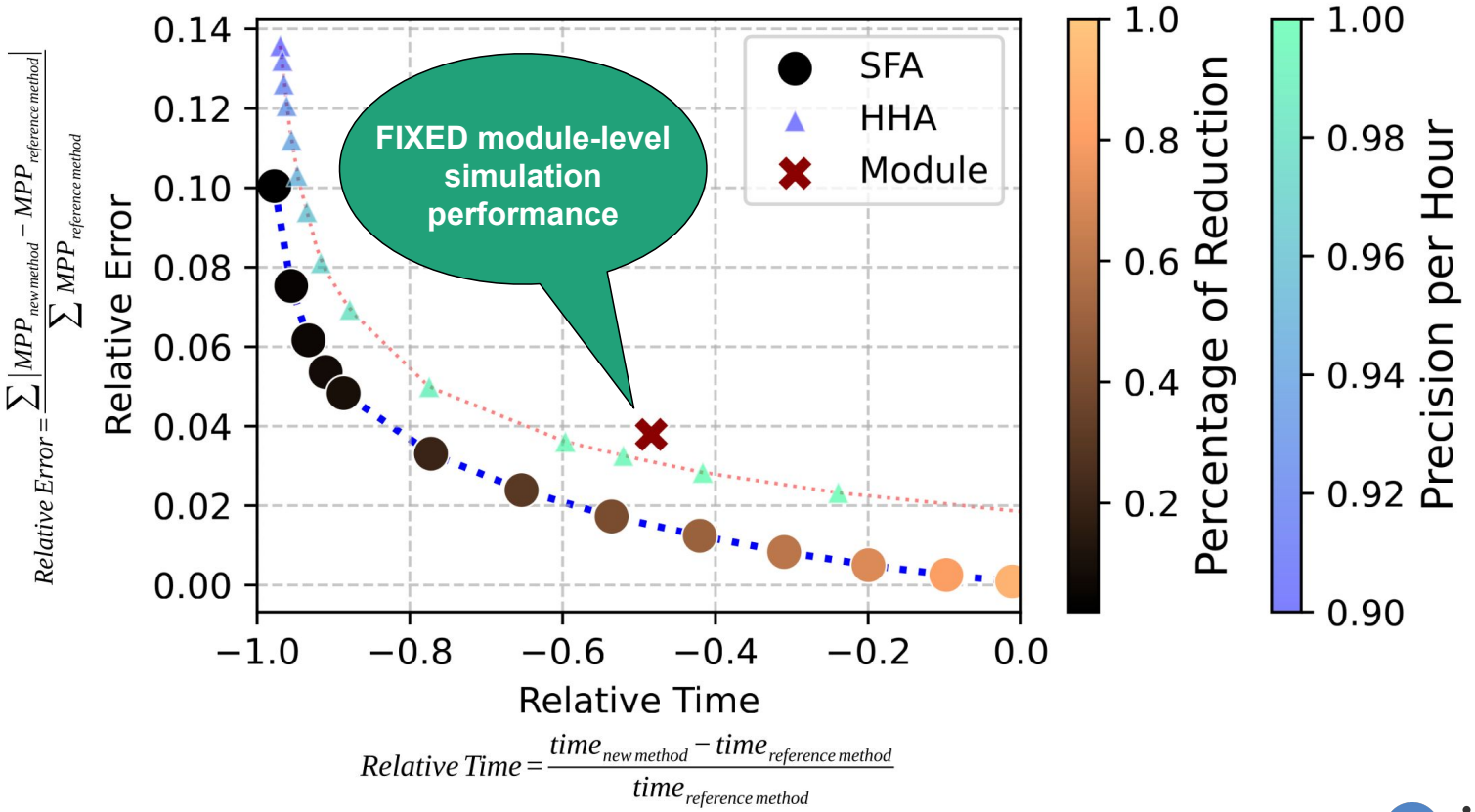
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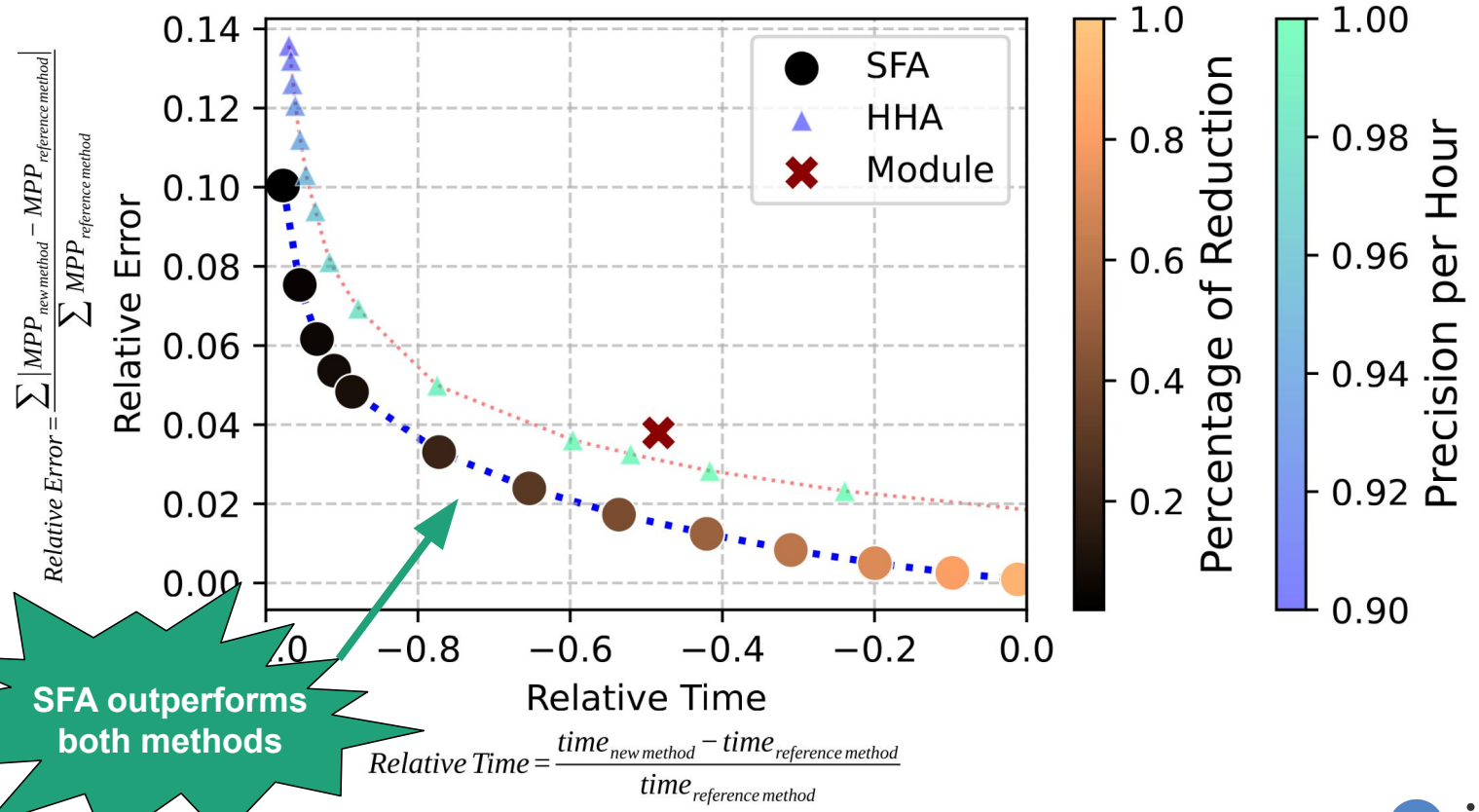
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Methods comparison (Pareto Frontier)



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SFA outperforms both methods

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Performance in terms of total annual power loss caused by partial shading (SFA)

$$MPP_{loss} = MPP_{no\,shading} - MPP_{shading}$$

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$$= \frac{\sum MPP_{loss\,new\,method}}{\sum MPP_{loss\,reference\,method}} - 1$$

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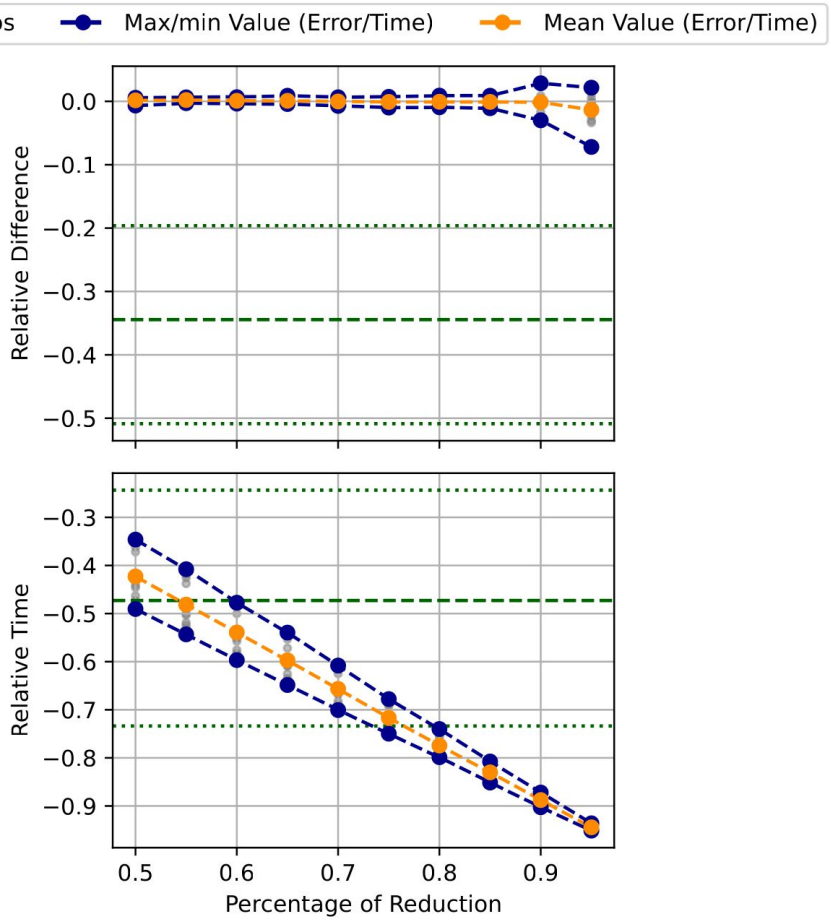
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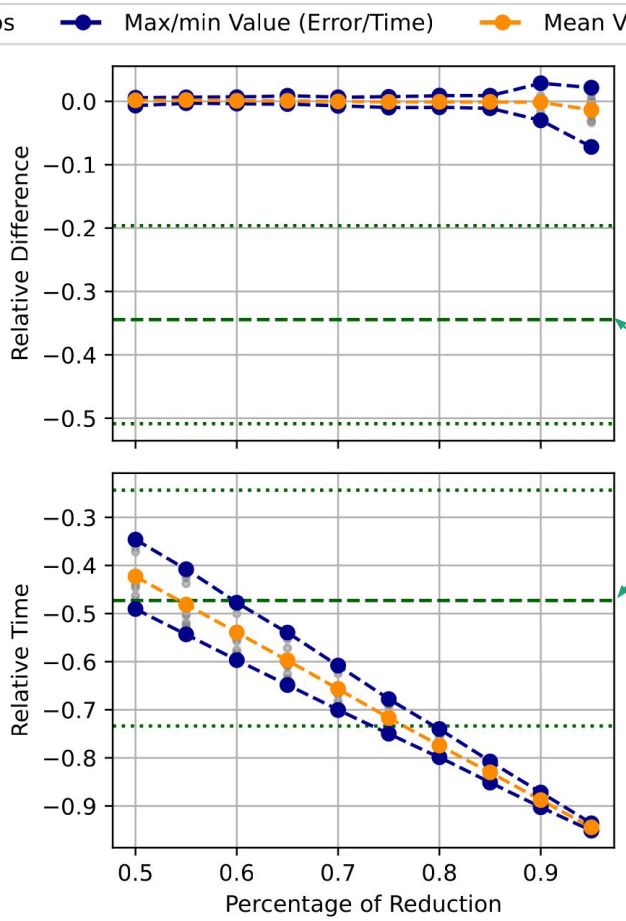
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Module-level performance for comparison

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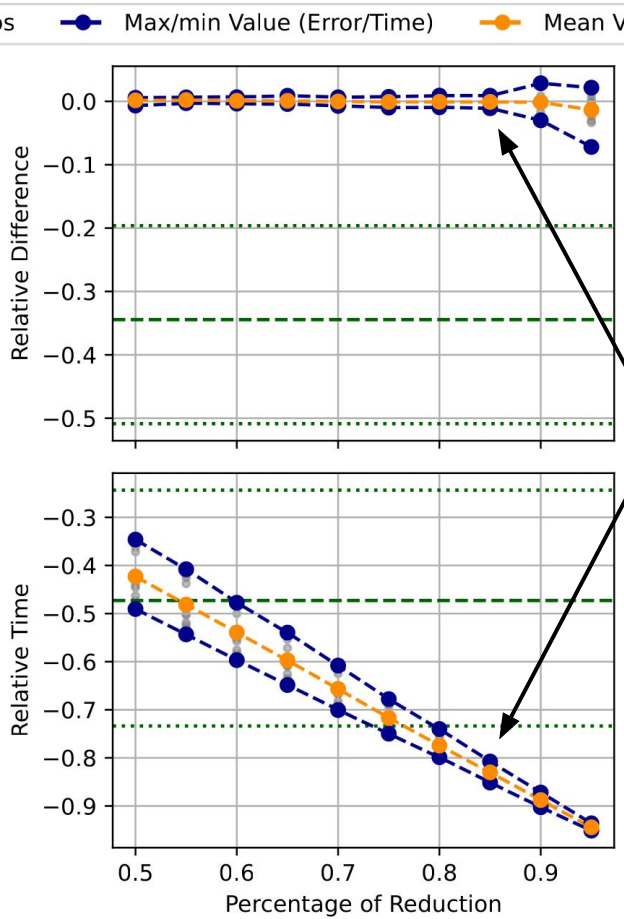
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Simulation 85% faster with nearly the same results!

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Summary

- Aggregation methods for initial data segmentation

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- Better alternative in terms of accuracy/speed trade-off than module-level simulation without aggregation

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- Aggregation methods for initial data segmentation
- Better alternative in terms of accuracy/speed trade-off than module-level simulation without aggregation
- Very tunable models depending on the use case

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END

PV shading models with pvlib

Accelerating photovoltaic system simulations via statistical data aggregation